SafeScape 2.0: An AI-Enhanced Gamified Intelligent Tutoring System for Fire Safety Preparedness Using Bayesian Knowledge Tracing, Dynamic Difficulty Adjustment, and Natural Language Processing

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BACHELOR OF SCIENCE IN COMPUTER SCIENCE

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APPROVAL SHEET

The thesis entitled "**SafeScape 2.0: An AI-Enhanced Gamified Intelligent Tutoring System for Fire Safety Preparedness Using Bayesian Knowledge Tracing, Dynamic Difficulty Adjustment, and Natural Language Processing**" prepared by **DELA PAZ, ZYRIL G., DELOS REYES, AXCEL ANDREI V., LACHANO JHED AERON V.** in partial fulfillment of the requirement for the degree of **BACHELOR OF SCIENCE IN COMPUTER SCIENCE**, major in **INTELLIGENT SYSTEMS**, is hereby recommended for approval and acceptance.

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INTRODUCTION

Fire safety is a major concern in the Philippines, yet its importance is often overlooked by the public until disaster strikes. The persistent threat is clear, with the Bureau of Fire Protection (BFP) reporting nine fire incidents in the National Capital Region on the first day of 2025 alone (Conejos, 2025). A significant portion of this problem stems from preventable hazards; BFP data shows that from January 1 to February 27, 2025, electrical issues were the leading cause of 625 fires nationwide (Rita, 2025). The BFP also points to other aggravating factors, linking the rise in fire incidents to extreme heat and fluctuations in electricity demand (Patron, 2025). This situation starkly demonstrates the limits of a reactive approach and points to the urgent need for better, community-focused education that shifts the focus from reacting to emergencies to proactively reducing risk.

In Sta. Cruz, Laguna, these risks are magnified by densely packed residential and commercial zones where a single localized fire can easily spread and endanger an entire neighborhood. While the local BFP conducts public education campaigns, the legacy iteration of the SafeScape platform in Sta. Cruz utilized a static Content Management System (CMS). While effective for basic information dissemination, it lacks pedagogical adaptability; users with varying experience levels receive the same linear educational flow, leading to suboptimal learning retention and instructional fatigue.

To modernize this infrastructure, SafeScape 2.0 transitions the platform into a Gamified Intelligent Tutoring System (ITS). The system utilizes a Closed-Loop Educational Machine Learning architecture designed to track, model, and dynamically respond to a user's specific knowledge gaps. To provide engaging cognitive training,

the system integrates a module called "The Right Call," an NLP-driven simulated fire-emergency communication and dispatch training module. This module uses a Prompt-Engineered Large Language Model (LLM) to process zero-shot or few-shot bilingual (Taglish) conversational inputs and dynamically assess procedural adherence.

Furthermore, to ensure the platform accommodates users with different familiarity levels, SafeScape 2.0 introduces two difficulty configurations within "The Right Call": Normal Mode and Professional Mode. Normal Mode utilizes structured multiple-choice options with contextual distractors to guide novice dispatchers, while Professional Mode utilizes typed, open-ended natural language inputs evaluated via Large Language Models. The interactive environment features a 2D patrol map of Sta. Cruz and live transcript scrolls designed to replicate a real emergency operations board. By combining Bayesian Knowledge Tracing (BKT) (Xu et al., 2022) with advanced Large Language Models and dynamic difficulty adjustment, SafeScape 2.0 aims to shift community safety from passive reading to active, measurable preparedness.

To appreciate the academic and technical significance of SafeScape 2.0 and "The Right Call," it is essential to examine the historical and legislative development of fire prevention in the Philippines. Under Presidential Decree No. 1185, enacted in 1977, and subsequently updated by Republic Act No. 9514 (otherwise known as the Fire Code of the Philippines of 2008), the Bureau of Fire Protection (BFP) was established as a distinct agency under the Department of the Interior and Local Government (DILG). The mandate of the BFP spans fire prevention, fire suppression, investigation, and enforcement of safety standards. However, despite these legislative guidelines, urban centers face unique systemic fire hazards. Rapid population growth,

illegal electrical connections, and poor enforcement of building codes in low-income neighborhoods contribute to a high risk of electrical and structural fires. Traditional public campaigns, which rely on pamphlets and annual fire drills during Fire Prevention Month in March, have struggled to adapt to modern media consumption habits. The public often perceives these campaigns as bureaucratic formalities, leading to low retention of critical emergency skills.

Furthermore, fire incidence data from the BFP indicates a rise in fires caused by electrical faults, such as substandard wiring, overloaded sockets, and poor appliance maintenance, particularly during summer heatwaves when the demand for domestic air conditioning and ventilation is highest. This structural and behavioral vulnerability underscores the need for educational programs that can reach a broad audience and maintain user engagement over time, transforming safety training from a passive compliance exercise into a practical, cognitive safety skill.

Modern Disaster Risk Reduction and Management (DRRM) frameworks, such as the Sendai Framework for Disaster Risk Reduction 2015-2030, advocate for decentralized, community-driven preparedness. This paradigm shift emphasizes that local communities are the first responders during any crisis. In Sta. Cruz, Laguna, local emergency response times are often slowed by traffic congestion, narrow, unmapped residential pathways (commonly referred to as eskinita), and geographical distances between rural barangays and the central fire station. Consequently, the initial minutes of an emergency are critical. The speed and accuracy with which a bystander or victim reports an incident to the BFP dispatch office determines the rescue team's deployment time and resource allocation. If a caller is unable to specify the barangay, the exact landmarks, or the material source of the fire, valuable minutes

are lost. Therefore, training citizens in effective emergency communication is not merely an educational exercise, but a high-stakes intervention.

In the context of local fire departments, dispatchers operate under extreme pressure, relying on structured intake protocols to extract critical information from highly distressed callers. "The Right Call" simulation directly addresses this operational bottleneck by placing community trainees in the role of dispatchers. This experiential roleplay forces users to appreciate the necessity of clear, structured communication. By experiencing the difficulty of extracting details from a panicked caller, players learn to identify the exact pieces of information—location, incident type, and active hazards—that they themselves must provide when reporting a real-life emergency.

Sta. Cruz, Laguna, serving as the provincial capital, presents a complex mix of commercial centers, historical municipal structures, and dense residential zones. Data obtained from the local Bureau of Fire Protection (BFP) station reveals that fire hazards vary significantly by barangay. In commercial hubs such as Barangay Bagumbayan and Barangay Poblacion, fire risks are heavily associated with high electrical loads in commercial spaces, overcrowded market stalls, and complex electrical service drops. In contrast, residential barangays such as Barangay Bubukal, Barangay Gatid, and Barangay Santisima Cruz present hazards linked to domestic activities, such as unattended cooking, faulty liquefied petroleum gas (LPG) regulators, and informal electrical connections (popularly termed "jumpers").

Furthermore, many of these residential zones are characterized by narrow streets and dense housing built from lightweight materials (Class A combustible fuels). The BFP response teams face challenges navigating these areas due to double-parked vehicles and low-hanging overhead cables. Consequently, the dispatch

simulation in SafeScape 2.0 incorporates these specific localized hazards, forcing the player to choose appropriate response units—such as traditional fire pumpers for Class A fires, chemical fire engines for industrial/chemical spills, or rescue squads for occupant extraction—based on the specific barangay map location and fire type described by the caller.

During an emergency, callers experience acute cognitive distress, which significantly degrades their communication capacity. Under the influence of high stress and fear, the human brain releases cortisol and adrenaline, triggering a "fight-or-flight" response that prioritizes basic survival mechanics over high-level cognitive processes. Psycholinguistic research indicates that under these conditions, callers experience:

1. Retrieval Deficits: Distressed individuals struggle to recall structured spatial information, such as street names or landmarks, relying instead on egocentric descriptions (e.g., "dito sa tapat ng kulay asul naming gate" - here in front of our blue gate) that are unhelpful to dispatchers unfamiliar with the immediate vicinity.

2. Speech Rate Shifts: Panicked callers often exhibit rapid, disorganized speech patterns or go into hyperventilation, causing them to stutter, omit crucial vowels, or speak in local dialects and colloquialisms without realizing they are not being understood.

3. Information Overload and Omission: Distressed individuals frequently focus on the immediate visual horror of the fire rather than answering the dispatcher's specific safety questions. For example, a caller might repeatedly scream "Ang laki ng apoy!" (The fire is so big!) while ignoring questions about active electrical hazards or trapped occupants.

"The Right Call" dispatch simulator replicates these cognitive challenges by utilizing Large Language Models to simulate panicked, realistic emergency dialogues. The trainee must learn to manage their own cognitive load, calm the virtual caller, and systematically extract the three vital components of an emergency report: the exact address, the nature of the fire (electrical, chemical, or structural), and the status of any occupants.

1.1 Research Problem

Despite ongoing fire safety awareness initiatives and the availability of digital learning platforms, existing systems—including the legacy SafeScape platform—remain largely static and generalized in their instructional approach. Most fire safety training applications adopt a one-size-fits-all design that lacks the capability to assess individual learner competencies, identify specific knowledge deficiencies, and adapt training content based on user performance. Consequently, learners receive uniform instructional experiences regardless of their varying levels of preparedness and understanding.

In rapidly urbanizing and densely populated communities such as Sta. Cruz, Laguna, inadequate personalization in fire safety education may contribute to persistent knowledge gaps, delayed emergency responses, and increased vulnerability to fire-related incidents. Furthermore, current systems rarely leverage artificial intelligence, machine learning, and adaptive simulation technologies to provide real-time competency assessment and individualized learning pathways. The absence of intelligent mechanisms for tracking learner progress, dynamically adjusting training difficulty, and simulating emergency communication scenarios limits the effectiveness of digital fire safety interventions.

To address these limitations, there is a need to develop an intelligent and adaptive fire safety training platform capable of continuously modeling learner knowledge, personalizing simulation experiences, and evaluating emergency response competencies through AI-driven interaction. Such a system may enhance learner engagement, improve knowledge retention, and strengthen community preparedness for fire emergencies.

Guided by these concerns, this study seeks to answer the following research questions:

1. How can a Knowledge Tracing model be designed and integrated into a web-based fire safety simulation game to identify, monitor, and predict individual learner knowledge gaps in fire prevention and emergency response?

2. How can a Dynamic Difficulty Adjustment (DDA) mechanism be developed to adapt simulation challenges based on learner performance and competency levels during gameplay?

3. To what extent does the implementation of adaptive learning mechanisms, including Knowledge Tracing and Dynamic Difficulty Adjustment, improve learners' fire safety knowledge, engagement, and emergency response performance compared to conventional static simulation environments?

4. How can a prompt-engineered Large Language Model (LLM) be utilized to generate realistic bilingual (Taglish) emergency communication scenarios and automatically assess user compliance with established emergency dispatch and reporting protocols?

5. What software architecture and integration strategies can be employed to effectively incorporate AI and machine learning microservices into the legacy

SafeScape PHP-based system while ensuring scalability, interoperability, maintainability, and real-time performance?

1.2 Research Objectives

The primary objective of this study is to design, develop, and evaluate SafeScape 2.0, a web-based Gamified Intelligent Tutoring System (ITS) that integrates Knowledge Tracing, Dynamic Difficulty Adjustment, and Large Language Models (LLMs) to deliver adaptive, personalized, and interactive fire safety training for the community of Sta. Cruz, Laguna.

Specifically, this study aims to:

1. Design and develop SafeScape 2.0, an AI-enhanced web-based Gamified Intelligent Tutoring System (ITS) that integrates fire safety assessment, simulation, and serious gaming modules to provide an interactive and adaptive learning environment for fire prevention, preparedness, and emergency response training.

2. Configure, calibrate, and evaluate a Bayesian Knowledge Tracing model for estimating learner mastery across defined fire-safety competencies coupled with a Dynamic Difficulty Adjustment (DDA) mechanism to predict learner mastery levels, identify knowledge deficiencies, and dynamically personalize simulation scenarios, hazard conditions, and learning pathways based on real-time user performance.

3. Design, implement, and experimentally evaluate a prompt-engineered Large Language Model (LLM)-based emergency dispatch simulator capable of generating realistic bilingual (Taglish) emergency scenarios and automatically assessing user adherence to established emergency communication and dispatch protocols.

4. Develop and optimize a hybrid software architecture that integrates machine learning microservices, LLM services, and game-based learning components with the legacy PHP-based SafeScape platform through API-driven communication and client-side state management, and evaluate its performance in terms of resource utilization and responsiveness under defined test conditions.

5. Experimentally determine the effectiveness of the proposed intelligent tutoring framework by comparing adaptive and non-adaptive learning environments, analyzing learning gains, knowledge retention, emergency response performance, user engagement, and system usability using appropriate statistical and software quality evaluation methods.

1.3 Research Framework

The research framework of SafeScape 2.0 is organized into two complementary parts: the theoretical framework and the conceptual framework. The theoretical framework identifies the established theories that explain the educational and adaptive principles underlying the study, while the conceptual framework translates these principles into the proposed system's inputs, processes, adaptive learning cycle, and intended outputs.

1.3.1 Theoretical Framework

The theoretical framework of SafeScape 2.0 is grounded in three complementary foundations: Intelligent Tutoring Systems (ITS), Bayesian Knowledge Tracing (BKT), and Cognitive Load Theory and Flow Theory. These foundations are organized hierarchically rather than treated as separate, unrelated theories. Intelligent Tutoring Systems provide the overall instructional structure; Bayesian Knowledge Tracing provides the learner-modeling basis; and Cognitive Load Theory and Flow Theory provide the pedagogical rationale for adapting task difficulty. In the proposed

system, Dynamic Difficulty Adjustment (DDA), gamification, Natural Language Processing (NLP), and Large Language Models (LLMs) are treated as implementation mechanisms that operationalize the theoretical foundations rather than as independent theoretical foundations.

1.3.1.1 Intelligent Tutoring Systems

Intelligent Tutoring Systems (ITS) provide the overarching theoretical foundation of SafeScape 2.0. An ITS is an educational system designed to provide individualized instruction by using information about learner performance to determine appropriate instructional actions. Unlike a fixed instructional sequence in which learners receive the same activities regardless of their performance, an ITS continuously considers the learner's current state and uses that information to guide subsequent learning activities.

The ITS structure adopted in the study consists of four major components: the Student Model, Domain Model, Pedagogical Model, and User Interface. The Student Model represents the learner's current knowledge or competency; the Domain Model represents the subject matter and skills to be learned; the Pedagogical Model determines appropriate instructional actions; and the User Interface provides the means through which the learner interacts with the system. This four-component structure is consistent with the ITS architecture described in the existing manuscript and literature base (Corbett & Anderson, 1994; Koedinger et al., 2012; Sottilare et al., 2018).

In SafeScape 2.0, the Student Model is operationalized through Bayesian Knowledge Tracing, which estimates learner mastery across defined fire-safety

competencies. The Domain Model contains the fire-safety knowledge, procedures, scenarios, and competencies used by the system. The Pedagogical Model is operationalized through adaptive instructional decisions, including Dynamic Difficulty Adjustment, which uses the learner's estimated competency to determine appropriate task demands. The User Interface consists of the interactive and gamified learning environment through which learners perform activities, respond to emergency scenarios, and receive feedback.

The ITS framework therefore establishes the overall relationship among learner state, domain knowledge, instructional decisions, and interaction. It provides the structure within which the other two theoretical foundations operate. SafeScape 2.0 is consequently not treated as a collection of unrelated artificial-intelligence features; rather, its components are organized as an adaptive tutoring process in which learner performance informs subsequent instructional actions.

1.3.1.2 Bayesian Knowledge Tracing

Within the ITS framework, Bayesian Knowledge Tracing (BKT) serves as the theoretical foundation for the Student Model. BKT represents learning as an evolving knowledge state that is inferred from observable learner performance. Because a learner's actual mastery cannot be directly observed during an interaction, BKT estimates the probability that the learner has mastered a particular skill based on the learner's previous responses.

The BKT model used in SafeScape 2.0 is based on the representation of learner knowledge as a latent state and learner responses as observable evidence. Its principal parameters are the probability of prior knowledge, probability of learning, probability of guessing, and probability of slipping. Following each relevant learner

interaction, the estimated mastery probability is updated according to the observed response. Repeated interactions therefore produce an evolving representation of the learner's knowledge state.

This theoretical role is important to SafeScape 2.0 because the system is intended to identify differences in learner competency rather than treat all learners as having the same level of preparedness. A learner may demonstrate stronger knowledge in evacuation procedures while requiring additional practice in electrical or chemical hazard recognition. BKT provides a principled way of representing these differences and supplying the adaptive instructional mechanism with an estimate of the learner's current mastery.

BKT is therefore positioned specifically as the learner-modeling foundation of the study. The mathematical implementation, parameter configuration, verification, and predictive validation of the model are addressed in the methodology and algorithm-analysis portions of the manuscript rather than being treated as part of the theoretical exposition.

1.3.1.3 Cognitive Load Theory and Flow Theory

Cognitive Load Theory (CLT) and Flow Theory provide the pedagogical foundation for determining how learning challenges should be presented to users. Cognitive Load Theory emphasizes the limited capacity of working memory and the importance of managing the amount and complexity of information that learners must process during a task. This is particularly relevant to fire-safety preparedness because emergency scenarios may require learners to process location information, hazard characteristics, response options, communication requirements, and time constraints within the same activity.

From the perspective of Cognitive Load Theory, SafeScape 2.0 should avoid introducing unnecessary task complexity before the learner has developed the prerequisite knowledge needed to process the activity effectively. The system therefore uses the learner's estimated competency as a basis for determining the level of support and complexity presented in subsequent activities. This principle is consistent with the manuscript's existing discussion of cognitive load in serious-game and emergency-training contexts.

Flow Theory complements Cognitive Load Theory by explaining the relationship between learner skill and task challenge. Sustained engagement is more likely when a challenge is appropriately matched to the learner's ability. Tasks that are substantially beyond the learner's capability may produce frustration or anxiety, whereas tasks that are too easy may produce boredom or disengagement. In SafeScape 2.0, this principle supports the progressive adjustment of scenario complexity and response demands as learner competency changes.

Together, Cognitive Load Theory and Flow Theory provide the pedagogical justification for adaptive difficulty. They explain why the system should avoid excessive cognitive demands for learners who have not yet developed sufficient mastery while progressively increasing challenges for learners who demonstrate stronger competency. Dynamic Difficulty Adjustment is therefore treated as an implementation mechanism that operationalizes these pedagogical principles, rather than as a separate theoretical foundation.

1.3.1.4 Synthesis of the Theoretical Framework

The three theoretical foundations form a hierarchical and interconnected basis for SafeScape 2.0. Intelligent Tutoring Systems provide the overall structure for

individualized learning. Bayesian Knowledge Tracing provides the learner-modeling foundation by estimating the learner's evolving mastery of fire-safety competencies. Cognitive Load Theory and Flow Theory provide the pedagogical rationale for adapting learning challenges according to the learner's demonstrated capability.

The resulting theoretical relationship can be expressed as follows: the learner interacts with an ITS-based learning environment; learner performance provides observable evidence; BKT uses this evidence to estimate the learner's knowledge state; the estimated state informs adaptive instructional decisions; and the adaptive mechanism adjusts subsequent learning activities in accordance with principles of manageable cognitive demand and appropriate challenge. The learner then produces new performance evidence, creating a continuous learning and adaptation cycle.

Within this structure, DDA translates learner-state information into changes in task difficulty, gamification provides the interactive learning context, and the LLM/NLP component supports natural-language emergency communication and response evaluation. These are consequently treated as implementation mechanisms rather than additional theoretical foundations. This organization keeps the theoretical framework focused on the principles that explain the proposed learning system while reserving detailed technical discussions for the Review of Related Literature and Research Methodology.

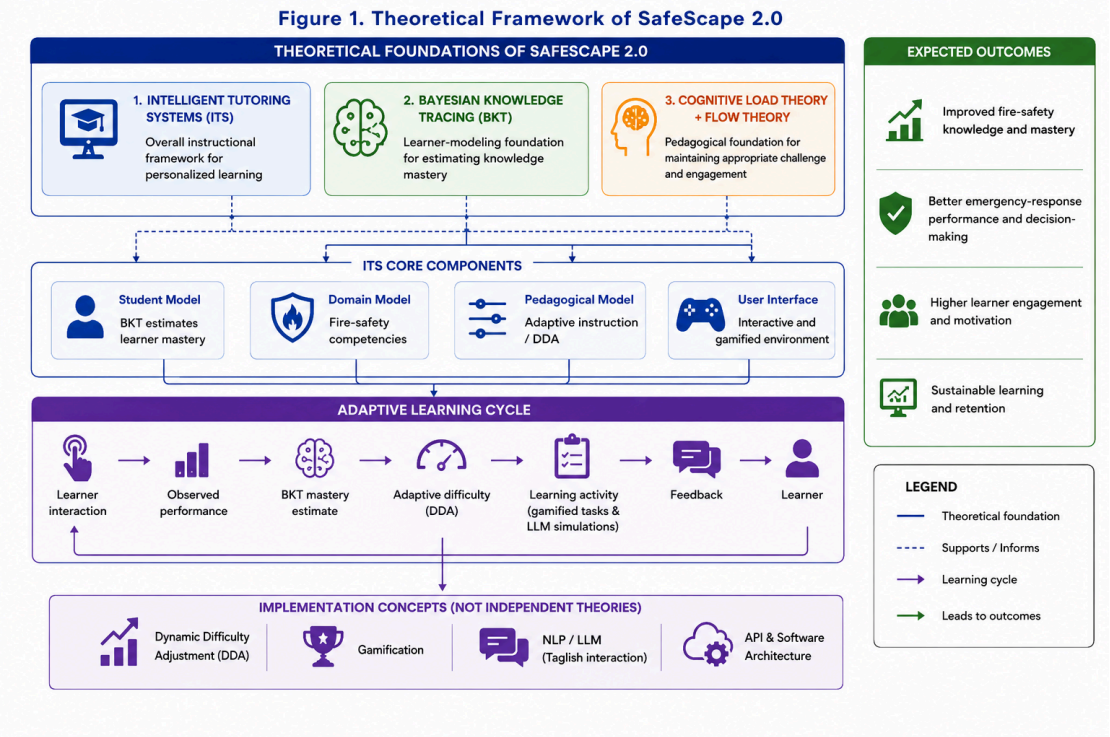


Figure 1. Theoretical Framework of SafeScape 2.0

1.3.2 Conceptual Framework

The conceptual framework translates the theoretical foundations of SafeScape 2.0 into the operational structure of the proposed system. It follows an Input-Process-Output (IPO) structure while incorporating a closed-loop adaptive learning cycle. The framework illustrates how learner information and learning interactions are processed by the Intelligent Tutoring System to estimate learner mastery, adapt task difficulty, deliver learning activities, and generate measurable educational and system outcomes.

Input

The input layer consists of the information required to establish and update the learner's learning state. These inputs include learner profiles, baseline or pre-test results, defined fire-safety competencies, learner responses, gameplay interactions,

and natural language inputs produced during emergency communication activities. Fire-safety knowledge and procedural requirements provide the domain content used to construct learning tasks and scenarios.

Process

The process layer is represented by the SafeScape 2.0 Intelligent Tutoring System. The Student Model uses Bayesian Knowledge Tracing to estimate the learner's mastery of targeted fire-safety competencies. Observable learner responses are treated as evidence for updating the estimated knowledge state. The Domain Model supplies the fire-safety competencies, procedures, scenarios, and instructional content that define what learners are expected to acquire.

The learner's estimated mastery is then used by the adaptive instructional mechanism. Dynamic Difficulty Adjustment modifies scenario difficulty according to the learner's demonstrated competency. Adaptation may involve changes in task complexity, time demands, environmental distractors, or other scenario conditions defined by the system. The pedagogical rationale for these changes is provided by Cognitive Load Theory and Flow Theory: learners should not be exposed to unnecessary cognitive demands before they are ready, while learners who demonstrate stronger mastery should receive appropriately challenging activities.

The adapted activities are delivered through the gamified learning environment and the emergency dispatch simulation, 'The Right Call.' The LLM/NLP component supports natural language interaction, including Taglish emergency communication, and provides automated assessment of learner responses against the defined communication and dispatch criteria. The LLM/NLP component is therefore

positioned as a system mechanism that supports the instructional process rather than as an independent learning theory.

Feedback is provided during and after learner interactions. Feedback may address response accuracy, safety or protocol compliance, and performance in the simulated communication task. The resulting performance information is returned to the Student Model, allowing the learner's estimated mastery to be updated and informing the next adaptive decision. This creates the closed-loop process central to the conceptual framework.

Output

The output layer consists of the intended learner and system outcomes of SafeScape 2.0. Learner-level outcomes include fire-safety knowledge gain, emergency-response performance, communication and protocol adherence, knowledge retention, and learner engagement. System-level outcomes include usability and user acceptance. These outcomes correspond to the study's stated objectives and provide the basis for evaluating the effectiveness of the adaptive learning environment.

The conceptual relationship begins with learner characteristics and baseline performance and proceeds through learner interaction with the ITS. Observable performance is processed by BKT to produce an updated estimate of learner mastery. The mastery estimate informs adaptive instructional decisions through DDA. The resulting learning activity provides another opportunity for the learner to practice the targeted competency. Performance from that activity is again incorporated into the learner model. The process therefore operates as a continuous feedback loop rather than a one-directional sequence.

The conceptual framework also distinguishes the adaptive learning mechanism from the technologies used to implement it. BKT represents the learner-modeling component; DDA represents the adaptive instructional mechanism; gamification provides the learning interaction context; and NLP/LLM technology supports the emergency communication module. Software architecture, API communication, local state management, and the Godot game environment support implementation but are not presented as theoretical variables.

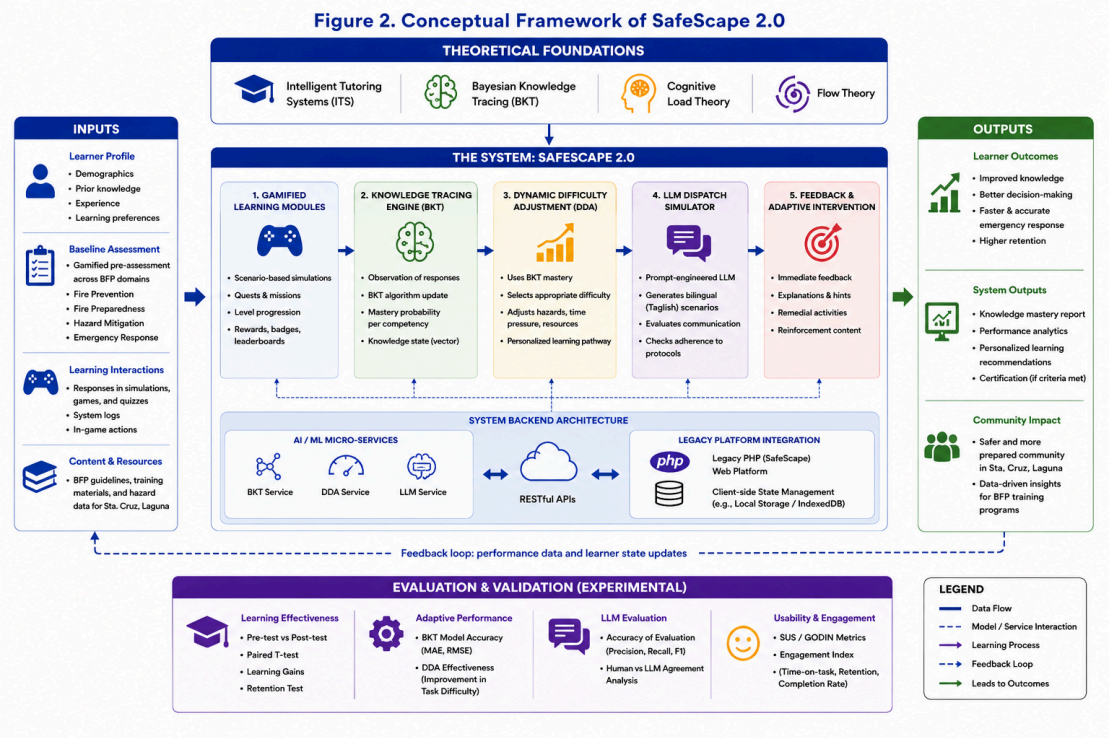


Figure 2. Conceptual Framework of SafeScape 2.0

1.4 Scope and Limitations of the Study

This study focuses on the comprehensive design, development, evaluation, and implementation of SafeScape 2.0, a web-based, gamified Intelligent Tutoring System (ITS) intricately tailored for proactive disaster preparedness and fire safety training.

The project, localized hazard mapping, and language models are specifically engineered for the municipality of Sta. Cruz, Laguna, which serves as the provincial

capital. The simulation deliberately incorporates the distinct spatial and structural vulnerabilities of this locale, contrasting the heavy electrical loads of commercial hubs like Barangay Bagumbayan and Barangay Poblacion with the domestic hazards and narrow pathways (*eskinita*) of residential zones like Barangay Bubukal, Barangay Gatid, and Barangay Santisima Cruz.

The system acts as a highly specialized Gamified ITS targeting official Bureau of Fire Protection (BFP) standard operating procedures. The educational matrix is strictly bounded to three critical safety domains: Electrical hazards (identifying overloaded circuits, Class C fires, and illegal connections), Chemical hazards (managing Class B/K grease fires and LPG leaks), and Evacuation protocols (crawling under smoke, using designated meeting points, and executing the P.A.S.S. extinguisher method).

The platform features advanced closed-loop machine learning microservices driven by a four-component ITS architecture: a Student Model, a Domain Model, a Pedagogical Model, and a User Interface. The core cognitive tracking is powered by Bayesian Knowledge Tracing (BKT), which mathematically models a learner's latent mastery across unlearned and learned states using Prior Knowledge, Learn Rates, Guess Rates, and Slip Rates. This feeds directly into a deterministic Dynamic Difficulty Adjustment (DDA) engine that procedurally scales scenario complexity, intake timers, and environmental distractors based on the user's real-time competency. The primary training module, "The Right Call," utilizes a prompt-engineered Large Language Model (specifically the gemini-3.1-flash-lite API) to simulate panicked emergency callers. It evaluates zero-shot and few-shot bilingual code-switching (Taglish) dispatch commands in real-time. To accommodate varying age groups, the

system offers a "Normal Mode" featuring structured multiple-choice options for novices, and a dialogue-driven "Professional Mode" for adults.

The standalone application is built entirely within the open-source Godot Engine (v4.x) utilizing native GDScript. The system relies on Client-Side State Management to serialize and persist user knowledge vectors locally directly to the device storage. It uses asynchronous JSON payloads formatted via HTTP Request nodes to communicate seamlessly with external machine learning microservices without interrupting the core game loop.

The overarching architecture is limited by the software engineering constraints of injecting advanced machine learning microservices and gamified WebGL components into the existing legacy SafeScape PHP/SQL web platform.

A primary limitation of the system is the inherent risk of the generative Large Language Model occasionally misinterpreting deep, unmapped regional slang or "hallucinating" incorrect emergency context and unscripted scenario elements. While the system employs strict prompt engineering boundaries, guardrails, and constrained JSON schema decoding to mitigate this, the AI is explicitly tuned for Taglish code-switching native to the Laguna region and may fail to accurately parse syntax from other distinct Philippine dialects.

The simulation's sophisticated conversational AI module in "Professional Mode" relies heavily on third-party API calls to external inference engines. Consequently, the system's responsiveness, Time-To-First-Token (TTFT) latency, and avoidance of frame-rate thread blockages are highly dependent on the stability of the municipal internet connection.

To completely prevent server-side bottlenecks and eliminate database latency during high-stress simulation loops, the system saves player profile data and BKT

matrices to local storage first. Data is synchronized to the central database asynchronously only upon module completion or when internet connectivity is detected, meaning persistent, real-time server tracking by administrators is limited by this localized, offline-first design.

1.5 Significance of the Study

This study represents a critical paradigm shift from passive information delivery to an active, decision-centric learning environment grounded in modern Disaster Risk Reduction and Management (DRRM) frameworks. The development of SafeScape 2.0 is expected to provide substantial, measurable benefits for the following groups:

1.5.1 Community Residents and Trainees

By utilizing an LLM for realistic response evaluation and Bayesian Knowledge Tracing for dynamic scenario generation, the platform creates an adaptive experience that scales perfectly beyond traditional one-size-fits-all training. It caters to everyone from young children identifying hazards in "Task Master" to adults navigating high-pressure procedural simulations in "The Right Call". The system directly addresses the severe psycholinguistic breakdowns and cognitive overload (e.g., retrieval deficits, rapid speech shifts) citizens experience during real emergencies, training them to remain calm, manage their mental load, and accurately communicate vital spatial and hazard data, thereby building genuine emergency response competency.

1.5.2 BFP Sta. Cruz Personnel

The system provides the Bureau of Fire Protection with a highly advanced, data-driven training analytics platform that modernizes their traditional public outreach campaigns. This allows fire officers to continuously monitor the exact

competency levels, specific knowledge gaps, and learning gains of the community. Furthermore, by effectively training civilians to act as competent first reporters who provide exact landmarks and hazard classes (e.g., Class A vs. Class C fires), the system directly helps mitigate operational bottlenecks, allowing dispatchers to deploy the correct emergency apparatus faster and more efficiently.

1.5.3 Schools and Educational Institutions

The platform provides highly interactive, flow-theory-based adaptive resources that can be deployed across local schools using existing low-spec hardware. It effectively replaces static manuals, pamphlets, and routine annual fire drills with intelligent tutoring, significantly reducing instructional fatigue and maximizing long-term knowledge retention among student trainees.

1.5.4 Future Researchers and Software Engineers

The project serves as a foundational architectural model for educational data mining and serious game development. It demonstrates highly effective integration strategies for injecting modern machine learning microservices—such as deterministic heuristic DDA and prompt-engineered LLMs—into lightweight game engines and legacy web platforms without requiring the immense computational overhead of training localized neural networks.

CHAPTER II

REVIEW OF RELATED LITERATURE

This review is organized by major constructs of SafeScape 2.0 rather than by isolated technologies. The synthesis progresses from the educational problem of fire safety, to game- and simulation-based learning, Intelligent Tutoring Systems and adaptive learning, Bayesian Knowledge Tracing, Dynamic Difficulty Adjustment, Cognitive Load and Flow, NLP/LLMs, hybrid integration, learning analytics, and evaluation.

2.1 Fire-Safety Education and Emergency Preparedness

Fire-safety education provides the domain foundation of SafeScape 2.0 because the system is intended to develop measurable knowledge and response competencies rather than simply disseminate safety information. Recent studies continue to show that structured instruction can improve fire-safety knowledge, attitudes, hazard recognition, and response preparedness. Yaman et al. (2024) examined teachers' fire-safety knowledge and training effectiveness, while Sethi et al. (2026) examined hospital staff preparedness. Mondal and Sinha (2025) similarly reported changes in knowledge and attitudes following formal fire-safety training. In occupational contexts, Smith et al. (2024) connected effective safety training with safety knowledge and protective behaviors. Taken together, these studies support the proposition that safety education is most useful when it develops actionable competencies rather than awareness alone.

A second pattern in the literature is the movement from conventional instruction toward interactive and simulated training. Carvalho et al. (2022) developed

a serious game for fire-evacuation procedures, while Daylamani-Zad et al. (2022) designed a serious game around decision-making under stress in a fire-evacuation scenario. Fu and Li (2023) reported the use of a virtual-reality serious game for fire-safety behavioral skills training, and Mutyam et al. (2026) described an AI-enhanced VR fire-evacuation training system for educational institutions. These studies demonstrate that digital environments can create opportunities to practice decisions and procedures without exposing learners to actual emergency risks.

The literature therefore establishes a clear educational need and a credible technological direction, but it also reveals a limitation relevant to SafeScape 2.0. Digital safety simulations can provide realistic scenarios, yet scenario exposure by itself does not necessarily provide a computational representation of how an individual learner's competency changes over time. SafeScape 2.0 addresses this limitation by defining fire-safety competencies as learner-model targets and using subsequent interaction evidence to update estimated mastery. Thus, the fire-safety literature justifies the domain and the use of practice-based scenarios, while the adaptive-learning literature provides the mechanisms for individualization.

2.2 Game-Based, Simulation, and Scenario-Based Learning

Recent evidence generally supports game-based learning as an effective instructional approach when game mechanics are meaningfully connected to learning objectives. Zhang and Yu (2022) distinguished the effects of gamification and game-based learning, reporting positive effects on learning achievement and motivation. Lei et al. (2022) found a positive effect of game-based learning on science achievement, while Hu et al. (2022) reported positive effects of game-based learning on cognition, retention, and motivation in chemistry. Lu et al. (2022) similarly reported positive effects on computational thinking. Barz et al. (2023) synthesized

digital game-based learning interventions and found positive effects across cognitive and affective-motivational outcomes. More recent reviews continue to report positive but context-dependent effects (Aster et al., 2024; Tene et al., 2025; Karimov et al., 2024).

Simulation and scenario-based learning are particularly relevant to SafeScape because fire emergencies require learners to interpret contextual information, choose actions, and experience consequences. Serious-game research in professional education suggests that realistic scenarios can support knowledge and skills development while reducing the risks associated with practicing in real environments (Wang et al., 2024; Wang et al., 2024; Facchino et al., 2025). High-stakes assessment research also emphasizes the importance of designing game tasks so that performance represents the intended competency rather than merely measuring game familiarity (Bijl et al., 2024). Emergency simulation research similarly indicates that simulated training can improve performance in time-critical tasks and non-technical skills such as communication, decision-making, and situation awareness (Abbas et al. (2023); Redjem et al. (2025); Pediatric simulation, 2022).

The literature also cautions against treating gamification as automatically beneficial. Reviews report that the effects of game elements vary according to implementation, learner characteristics, instructional context, and the specific mechanics used (Zeybek & Saygı, 2024; Dehghanzadeh et al., 2024; Faísca et al., 2025; Ramírez Ruiz et al., 2024). This supports the position adopted in the revised theoretical framework: gamification is a learning-design mechanism, not an independent theoretical foundation. In SafeScape 2.0, game mechanics provide the environment in which learners practice fire-safety competencies, while the adaptive

mechanisms determine how the instructional challenge responds to learner performance.

2.3 Intelligent Tutoring Systems and Adaptive/Personalized Learning

Adaptive and personalized learning research establishes the need to move beyond fixed instructional sequences. Recent systematic reviews describe adaptive systems as environments that use learner information to modify content, feedback, sequence, or difficulty. Bayly-Castaneda et al. (2024) reviewed AI-mediated personalized learning paths and found a strong emphasis on adaptive learning technologies. Hariyanto et al. (2025) likewise identified AI techniques that enable data-driven personalization and real-time adaptation. A scoping review of personalized adaptive learning in higher education reported improvements in academic performance in a substantial proportion of reviewed studies, while also identifying resource and implementation constraints (Personalized adaptive learning scoping review, 2024).

Intelligent Tutoring Systems provide a natural architecture for these adaptive functions because they organize learner information, domain knowledge, instructional decisions, and interaction into a coherent tutoring process. Recent reviews of ITS research report continued growth in AI-supported tutoring and personalized instruction (Cuéllar-Rojas et al., 2022; Lin et al., 2023; Létourneau et al., 2025; Romano et al., 2025; Rodrigues et al., 2025). Létourneau et al. (2025), for example, synthesized 28 studies involving 4,597 K–12 learners and found generally positive effects of AI-driven ITS on learning and performance, while also noting the need for longer interventions, larger samples, and attention to ethical issues.

For SafeScape 2.0, the literature supports an ITS-level organization rather than treating BKT, DDA, and LLMs as unrelated technologies. The learner model supplies information about the learner's current state; the domain model supplies fire-safety competencies; the pedagogical mechanism uses the learner state to determine subsequent learning activities; and the interface delivers those activities. Personalization is consequently implemented through the interaction of these components. This directly aligns the RRL with the revised theoretical framework in which ITS is the overarching foundation.

2.4 Bayesian Knowledge Tracing and Probabilistic Learner Modeling

Knowledge Tracing research focuses on estimating a learner's evolving knowledge state from sequential interactions. Recent surveys emphasize that knowledge tracing supports personalized instruction by allowing educational systems to infer latent mastery and predict subsequent performance (Abdelrahman et al., 2023). Bulut et al. (2023) demonstrated the accessibility of BKT through pyBKT and described the model's use of probabilistic parameters to update mastery. Chan et al. (2022) examined multi-skill learner performance using Bayesian approaches, while Shchepakin et al. (2024) investigated constraints for BKT parameters. These studies reinforce the value of interpretable probabilistic learner models for adaptive educational environments.

Recent work also demonstrates the broader knowledge-tracing landscape. Minn et al. (2022) proposed interpretable knowledge tracing that emphasizes skill mastery and explainability, while Sun et al. (2022) explored ensemble knowledge tracing for modeling interactions. Zhang et al. (2022), Park et al. (2022), and Wang et al. (2023) illustrate how researchers continue to extend learner models. More recent work includes Zhang et al. (2024), Wang et al. (2025), and Shimada and Okada

(2025). These studies are useful for positioning standard BKT as an interpretable baseline within a broader family of learner models.

For SafeScape 2.0, BKT is appropriate because the system requires skill-level mastery estimates that can be updated after learner responses and interpreted by a downstream difficulty-selection mechanism. The model's probabilistic nature also prevents the system from treating a single correct answer as conclusive evidence of mastery or a single incorrect answer as conclusive evidence of non-mastery. The literature therefore supports BKT as the learner-modeling foundation, while more complex knowledge-tracing approaches provide comparison and context rather than requiring SafeScape to adopt a deep neural learner model.

2.5 Dynamic Difficulty Adjustment and Adaptive Challenge

Dynamic Difficulty Adjustment refers to changing task demands in response to learner performance or estimated capability. Recent research demonstrates that adaptive difficulty can be implemented using deterministic rules, fuzzy systems, reinforcement learning, or other computational approaches. Chrysafiadi et al. (2023) presented a fuzzy-based DDA mechanism for an educational 3D game, while Shohieb et al. (2022) reported higher engagement under DDA than under classical level advancement. Seyderhelm and Blackmore (2024) investigated dynamic adaptive training using performance and cognitive-load indicators, providing evidence that challenge can be adjusted according to learner state.

More recent work extends DDA toward AI-driven adaptive learning and reinforcement-learning approaches (Skaiky et al., 2025; Ruan & Lu, 2025). A 2026 literature review of DDA in serious games further indicates that difficulty adaptation has been applied across different serious-game contexts and that the choice of

adaptation strategy depends on the type of learner evidence, game, and instructional objective (Víteková et al. (2026)). These findings suggest that there is no single universal DDA method. Instead, adaptation should be selected according to the information available about learners and the requirements of the learning environment.

SafeScape 2.0 uses a deterministic, rule-based DDA mechanism in which BKT mastery estimates are mapped to predefined difficulty levels. This choice emphasizes interpretability and predictable behavior in a safety-training context. Importantly, DDA is not treated as an additional educational theory. It is the mechanism through which the pedagogical rationale from Cognitive Load Theory and Flow Theory is operationalized. The system therefore uses learner mastery as evidence, DDA rules as the adaptation mechanism, and task difficulty as the instructional response.

2.6 Cognitive Load, Flow, Engagement, and Adaptive Learning

The revised theoretical framework identifies Cognitive Load Theory and Flow Theory as the pedagogical foundations for adaptive challenge. Recent research supports the importance of managing task demands in interactive learning. The randomized study of adaptive scaffolding in game-based learning reported relationships among adaptive support, performance, cognitive load, and engagement, illustrating how learner-state information can be used to adjust instructional support (Faber et al. (2024)). Research examining the cognitive cost of serious-game interfaces similarly shows that interface complexity can impose additional load and potentially reduce intervention effectiveness (Bu et al. (2025)).

Flow-related research emphasizes the relationship between challenge and learner capability. In a randomized serious-game study, facilitation that moderated challenge levels was associated with stronger flow outcomes and positive learner perceptions (Tan et al. (2024)). Serious-game reviews also identify challenge, feedback, and progression as important design features that influence engagement and learning (Aster et al., 2024; Karimov et al., 2024). These findings support the idea that adaptive challenge should not be understood simply as making a game harder. It should be adjusted so that the learner is neither overwhelmed nor consistently under-challenged.

For SafeScape 2.0, Cognitive Load Theory and Flow Theory therefore explain why DDA is pedagogically useful. Lower mastery estimates can justify simpler tasks, greater support, or lower information density, while stronger mastery estimates can justify more complex hazards, shorter decision windows, or more demanding communication scenarios. This relationship is intentionally narrower than the earlier framework: scaffolding and feedback remain design strategies, while CLT and Flow provide the principal pedagogical justification for adaptive difficulty.

2.7 Artificial Intelligence, NLP, and Large Language Models in Education

Artificial Intelligence in Education has expanded from predictive and rule-based systems toward generative and language-based tools. Recent reviews identify personalization, tutoring, assessment, feedback, and content generation as major application areas (Wang et al. AIED, 2024; Xu et al., 2024; Hariyanto et al., 2025). Large Language Models became particularly prominent after the release of ChatGPT, leading to a rapid increase in empirical educational research. Reviews of ChatGPT and LLM applications identify tutoring, feedback, assessment, content

generation, and learning support as recurring applications (Albadarin et al., 2024; Montenegro-Rueda et al., 2023; Baig & Yadegaridehkordi, 2024; Qian, 2025).

Evidence concerning learning effects is encouraging but not uniformly positive. Deng et al. (2025) synthesized experimental studies and reported positive effects on academic performance, affective-motivational outcomes, and higher-order thinking, while also emphasizing methodological limitations. Lo et al. (2024) reported mixed engagement effects across 72 studies. Jin and Sercu (2025) found that ChatGPT interventions were generally effective for knowledge acquisition but more variable for skills development. Research at K–12 level similarly identifies opportunities alongside concerns about reliability, over-reliance, privacy, and implementation (Zhang, 2024; Marzano, 2025; Alfarwan, 2025).

These findings support the use of an LLM in SafeScape 2.0 as an interaction and evaluation mechanism, but they do not justify treating LLMs as a separate educational theory. The system's LLM component processes natural-language emergency responses and returns structured evaluations against predefined criteria. The literature therefore informs prompt design, output constraints, reliability testing, and human validation rather than defining the core theoretical foundation of the tutoring system.

2.8 LLM-Based Assessment, Feedback, Reliability, and Structured Output

Automated assessment of open-ended language is a major application of NLP and LLMs in education. Sadanand et al. (2022) demonstrated automated essay evaluation using NLP and sentiment features. Wangwiwattana and Tongvivat (2023) examined an LLM-based academic assessment approach, while Caines et al. (2023) discussed the opportunities and challenges of LLMs for teaching and assessment

technology. More recent systematic evidence shows substantial interest in LLM-powered automated assessment, but also emphasizes the need for reliable evaluation and human reference standards (Emirtekin, 2025; Gao et al. (2024)).

Reliability is especially important when AI outputs are used to judge learner performance. Seo et al. (2025) examined consistency and accuracy of LLM evaluators in education, while Mendonça et al. (2025) compared LLM scoring with formative-assessment evaluation. Explainability has also become a concern, with Li et al. (2024) exploring automated educational assessment that provides rationale alongside scores. Hallucination research further demonstrates why generated output cannot automatically be assumed to be correct (Li et al., 2023; Zhu et al., 2024).

SafeScape 2.0 consequently requires structured outputs, explicit criteria, and independent validation. Research on structured generation includes schema-based approaches and post-processing methods for producing machine-readable outputs (Lu et al., 2025; Wang et al., 2025; Shorten et al., 2024). In the proposed system, these ideas support the use of constrained JSON-like outputs and predefined communication criteria. Precision, recall, F1-score, and agreement with domain experts can then be used to evaluate whether the LLM's categorical judgments are sufficiently reliable for educational use.

2.9 Hybrid AI Integration and Learning Analytics

The SafeScape architecture combines computational mechanisms that perform different functions rather than relying on one monolithic model. Current adaptive-learning research increasingly describes combinations of learner modeling, AI recommendation, generative AI, and real-time analytics (Bayly-Castaneda et al., 2024; Hariyanto et al., 2025; Iqbal & Kene, 2026). The literature suggests that

personalization can benefit from multiple data sources and computational mechanisms, but integration also introduces concerns about interoperability, transparency, reliability, and system complexity.

Learning analytics provides the data layer connecting learner behavior to adaptive decisions. Reviews of learning analytics describe the use of interaction traces, performance data, dashboards, feedback, and predictive models to understand learner progress (Mian et al., 2022; Heikkinen et al., 2023; Palanci et al., 2024; Paulsen & Lindsay, 2024). Human-centered learning-analytics research additionally emphasizes privacy, human control, trust, and transparency (Khalil et al., 2023; Cabral et al., 2025). In immersive environments, analytics can also be combined with interaction traces to study learner behavior and educational outcomes (Sakr & Abdullah, 2024).

For SafeScape 2.0, interaction logging is therefore not merely a technical storage function. Learner responses, scores, timestamps, attempts, scenario outcomes, and difficulty transitions provide evidence for BKT updates and later evaluation. The architecture can consequently be represented as a modular pipeline: learner response → BKT update → mastery estimate → DDA decision → activity selection, while the LLM pathway separately processes language responses and produces structured feedback. This separation improves interpretability because each component has a defined computational role.

2.10 Evaluation of Adaptive Intelligent Learning Systems

Evaluation literature indicates that adaptive educational systems should be evaluated at multiple levels rather than only by whether the software runs. Learning outcomes can be examined through baseline and post-intervention measures, while

interaction data can provide evidence of performance during the intervention. Serious-game assessment literature emphasizes the need to align game performance measures with the intended competency (Bijl et al., 2024). Reviews of serious games and simulation also emphasize the importance of rigorous experimental designs and longer-term outcome measures when evaluating educational effectiveness (Tene et al., 2025; Wang et al., 2024; Redjem et al. (2025)).

Usability and acceptance provide a complementary perspective. Recent usability reviews demonstrate that standardized questionnaires remain widely used in educational technology evaluation, including SUS and PSSUQ/CSUQ (Vlachogianni & Tselios, 2022; Vlachogianni and Tselios (2023)). Acceptance research indicates that perceived usefulness, trust, effort expectancy, and attitudes can influence willingness to use AI systems (Kelly et al., 2023; Fang et al., 2024; Ali et al., 2025). Studies of AI acceptance in education further suggest that AI literacy and contextual factors can shape acceptance (Al-Momani & Ramayah, 2024; Nevárez Montes et al., 2025; Salhab & Aboushi, 2025).

For SafeScape 2.0, these findings justify a multi-dimensional evaluation framework covering learning gain, adaptive behavior, LLM/NLP validity, usability, user acceptance, and technical performance. The RRL should not prescribe the exact statistical procedure in Chapter II; instead, it establishes why these dimensions are appropriate. The detailed pre-test/post-test design, BKT verification, DDA functional testing, LLM expert validation, usability testing, and system performance testing should remain in Chapter III.

2.11 Synthesis and Research Gap

Across the reviewed literature, several conclusions converge. First, fire-safety and emergency training benefits from structured instruction and opportunities to practice realistic decisions (Carvalho et al., 2022; Daylamani-Zad et al., 2022; Fu & Li, 2023; Yaman et al., 2024; Mondal & Sinha, 2025). Second, game-based, serious-game, and simulation-based approaches can support learning, engagement, and procedural practice when their mechanics are aligned with instructional goals (Lei et al., 2022; Zhang & Yu, 2022; Barz et al., 2023; Aster et al., 2024; Tene et al., 2025). Third, adaptive-learning and ITS research demonstrates the value of using learner information to personalize instruction (Lin et al., 2023; Bayly-Castaneda et al., 2024; Hariyanto et al., 2025; Létourneau et al., 2025).

Fourth, knowledge-tracing research provides computational approaches for estimating evolving learner states, with BKT remaining an interpretable probabilistic option for sequential skill modeling (Bulut et al., 2023; Abdelrahman et al., 2023; Shchepakin et al., 2024; Pradhan et al., 2026). Fifth, DDA research demonstrates that task difficulty can be adapted according to learner evidence, while cognitive-load and flow research provides a pedagogical rationale for matching challenge to learner capability (Chrysafiadi et al., 2023; Seyderhelm & Blackmore, 2024; Faber et al. (2024)). Sixth, LLM/NLP research shows promise for language-based feedback and assessment but also establishes the need for structured outputs, reliability checks, and human validation (Emirtekin, 2025; Seo et al., 2025; Li et al., 2023; Zhu et al., 2024).

The resulting gap is therefore not that fire-safety games, adaptive learning, BKT, DDA, or LLMs have never been studied. Rather, the identified gap concerns their integration and evaluation in a single, competency-oriented fire-safety learning environment. Few identified studies combine a probabilistic learner model, rule-based

adaptive difficulty, and language-model-supported emergency communication within a gamified fire-safety tutoring system. SafeScape 2.0 addresses this gap by integrating BKT for learner mastery estimation, DDA for adaptive scenario difficulty, and NLP/LLM functionality for emergency communication and response evaluation within an ITS-based game environment. This integration is intended to move fire-safety education from predominantly fixed content delivery toward measurable, adaptive competency development.

The literature reviewed across the major themes converges on a common direction: effective fire-safety education benefits from opportunities to practice realistic decisions, while digital games and simulations provide controlled environments in which such practice can occur. Recent fire-safety studies demonstrate that serious games and virtual environments can improve knowledge, evacuation behavior, and behavioral skills, but these environments do not inherently maintain an explicit computational representation of each learner's evolving competency.

Research on Intelligent Tutoring Systems and adaptive learning addresses this limitation by using learner information to individualize instructional content and sequence. Within this literature, Bayesian Knowledge Tracing remains relevant because it produces interpretable probabilistic estimates of skill mastery from sequential learner responses. Dynamic Difficulty Adjustment then provides a practical mechanism for converting learner-state information into changes in task challenge. Recent DDA research further shows that adaptation can be based on performance, cognitive load, or other learner indicators, although there is no single adaptation strategy that is optimal across all serious-game contexts.

Cognitive Load Theory and Flow Theory provide the pedagogical rationale for this adaptation. Recent experimental research indicates that adaptive support and challenge can influence cognitive load, performance, and engagement, while serious-game studies demonstrate that inappropriate challenge or interface complexity can undermine the learning experience. Accordingly, SafeScape 2.0 treats DDA as an implementation mechanism supported by these pedagogical principles rather than as an independent educational theory.

Recent Artificial Intelligence and LLM literature provides an additional technological opportunity for natural-language interaction and automated response assessment. Systematic reviews show that LLMs can support tutoring, feedback, personalization, and automated assessment, but they also identify concerns involving hallucination, inconsistency, fairness, transparency, privacy, and over-reliance. These findings support the use of structured outputs, explicit evaluation criteria, and independent expert validation in SafeScape 2.0.

Taken together, the literature does not indicate that any single component—fire-safety simulation, game-based learning, BKT, DDA, or LLM assessment—is itself novel. The more specific gap concerns their integration and evaluation within one competency-oriented fire-safety tutoring environment. Few identified studies combine a probabilistic learner model, rule-based adaptive difficulty, and language-model-supported emergency communication and response evaluation within a gamified fire-safety Intelligent Tutoring System. SafeScape 2.0 addresses this integration gap by using BKT for learner mastery estimation, DDA for adaptive scenario difficulty, and NLP/LLM functionality for natural-language emergency communication and structured response evaluation.

CHAPTER III

RESEARCH METHODOLOGY

This chapter presents the materials and methods used in the design, development, implementation, and evaluation of SafeScape 2.0, an adaptive gamified intelligent tutoring system for fire-safety preparedness. It describes the research design, study locale, applied computing concepts, algorithm analysis, data collection procedures, data-model generation, system development methodology, software tools, system architecture, and software testing and validation procedures.

The methodology is designed to address both the software-development requirements and the research-validation requirements of the study. Software verification determines whether the system functions according to its specifications, while research validation determines whether the implemented BKT, DDA, and LLM/NLP components produce valid and measurable outputs.

3.1 Research Design

The study will employ a developmental and evaluative research design with a quasi-experimental component. The developmental component will be used to design, implement, integrate, and test SafeScape 2.0, while the evaluative component will determine the technical quality, usability, adaptive behavior, and educational effectiveness of the developed system.

The study will combine development and application research with an experimental evaluation of the system's adaptive learning mechanisms. This approach is appropriate for a computer science study because the research problem requires

both the construction of a functional software artifact and empirical evaluation of the computational techniques incorporated into the artifact.

The developmental component will focus on the design, development, and integration of the major components of SafeScape 2.0, including a gamified Intelligent Tutoring System (ITS) for fire-safety preparedness, Bayesian Knowledge Tracing (BKT) for estimating and maintaining learner mastery, Dynamic Difficulty Adjustment (DDA) for adapting scenario difficulty based on learner performance, and Natural Language Processing (NLP) through a Large Language Model (LLM) for evaluating emergency communication. The system will also incorporate learner-profile persistence to preserve gameplay progress and knowledge states across sessions, together with interactive emergency scenarios that provide opportunities for learners to practice fire-safety decision-making and communication in simulated situations.

The evaluative component will assess SafeScape 2.0 across four major dimensions: algorithmic validity, focusing on the correctness and predictive performance of the Bayesian Knowledge Tracing (BKT) model; adaptive effectiveness, examining the proper operation of the Dynamic Difficulty Adjustment (DDA) mechanism and its contribution to learner performance; LLM/NLP performance, determining the accuracy and agreement of system-generated emergency-response evaluations with assessments provided by domain experts; and system and educational quality, encompassing functionality, usability, performance, user acceptance, and learning outcomes.

3.1.1 Quasi-Experimental Component

To evaluate whether adaptive learning provides an advantage over non-adaptive learning, the study will employ a pre-test/post-test comparison design. Where participant allocation is feasible, participants will be assigned to an adaptive and a non-adaptive condition.

Table X. Pre-test–Post-test Comparison Group

Group	Pre-test	Intervention	Post-test
Experimental Group	O_1	SafeScape 2.0 with BKT + DDA	O_2
Comparison Group	O_3	Non-adaptive/static difficulty condition	O_4

The experimental group will use the adaptive version of SafeScape 2.0 in which learner mastery estimates generated by BKT influence the DDA mechanism. The comparison group will use a version with predetermined or fixed difficulty levels. Both groups will receive equivalent instructional content and fire-safety scenarios to ensure that the primary difference between the conditions is the adaptive mechanism.

Pre-test and post-test scores will be used to determine changes in fire-safety knowledge. Where the assumptions of the selected statistical test are satisfied, inferential analysis will be conducted at a 0.05 level of significance. The effect size will also be reported to determine the practical magnitude of the observed difference.

If a delayed assessment is included, a retention test will be administered after the intervention to determine whether learning is maintained beyond the immediate post-test.

3.1.2 Research Validation Components

Separate validation activities will be conducted for the major intelligent components of SafeScape 2.0.

Table 2. Research Components and Validation Measures

Component	Validation Focus	Main Evidence
BKT	Mathematical correctness and predictive validity	Expected outputs, prediction metrics
DDA	Rule correctness and adaptive effectiveness	Difficulty transitions, learning outcomes
LLM/NLP	Taglish interpretation and protocol evaluation	Expert agreement, classification metrics
System	Functionality and quality	Functional and performance tests
Usability	User experience	SUS and task-based evaluation
Learning	Knowledge improvement	Pre-test/post-test results

This table presents the major components of SafeScape 2.0 and their corresponding validation focus and sources of evidence. Each component is evaluated using measures appropriate to its intended function, ensuring that the assessment

covers not only system functionality but also the validity of the BKT and DDA mechanisms, the performance of the LLM/NLP component, usability, and learning outcomes.

3.2 Locale of the Study

The study will be conducted in **Santa Cruz, Laguna**, with the fire-safety and disaster-preparedness context anchored on the operational environment of the **Bureau of Fire Protection (BFP) Santa Cruz Station** and the communities served by the municipality.

Santa Cruz serves as an appropriate locale because the study is designed to develop a localized fire-safety learning environment reflecting community-level emergency situations. The locale provides the contextual basis for identifying relevant fire hazards, emergency communication requirements, spatial references, and preparedness practices that can be incorporated into the simulation.

The study will consider fire-safety scenarios relevant to both residential and commercial environments. These may include electrical fires, cooking-related incidents, LPG-related hazards, combustible materials, evacuation situations, and emergency communication scenarios.

The locale will provide the contextual and operational basis for the The Right Call module, in which learners will engage in simulated fire-related emergency communication scenarios representative of the local setting. The module will require participants to communicate essential incident information, including the precise location, identifiable landmarks, nature and type of emergency, apparent hazards, presence of victims or potentially trapped individuals, immediate threats, and other

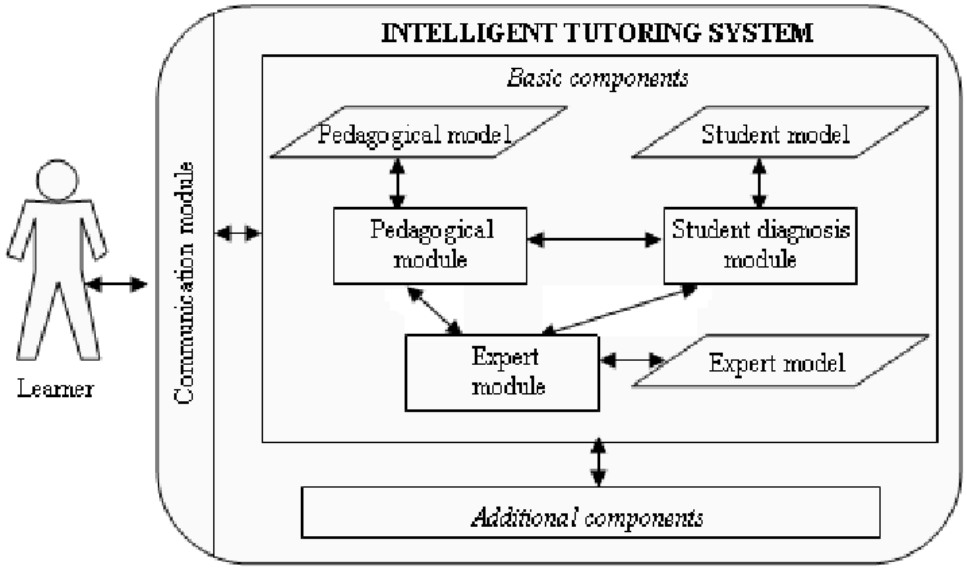
information relevant to emergency assessment and response. These variables are intended to represent the critical information required for structured emergency communication and to provide measurable indicators of the learner's ability to convey accurate, complete, and relevant information under simulated emergency conditions. By incorporating localized scenarios, the module provides a controlled environment for assessing and developing emergency communication and decision-making competencies without exposing participants to actual emergency hazards. Domain-specific information obtained from qualified fire-safety personnel will be used for content validation, scenario development, verification of emergency protocols, and validation of the LLM-based evaluation criteria. Such expert-derived information will serve as the basis for establishing the content and evaluation standards, while learner performance data will constitute the primary empirical evidence for evaluating the educational effectiveness of the system.

3.3 Applied Concepts and Techniques

SafeScape 2.0 applies concepts from Artificial Intelligence, Intelligent Tutoring Systems, probabilistic learner modeling, adaptive learning, Natural Language Processing, gamification, and software engineering.

3.3.1 Intelligent Tutoring System

SafeScape 2.0 adopts an Intelligent Tutoring System (ITS) architecture to support individualized and adaptive fire-safety learning. Unlike conventional instructional systems that provide uniform learning activities, the ITS continuously represents the learner's evolving competency through a learner model and uses the resulting knowledge state to inform subsequent instructional decisions.



https://www.researchgate.net/figure/The-general-architecture-of-an-intelligent-tutoring-system-adopted-from-28_fig1_240700291

Figure 3.X. Adapted Intelligent Tutoring System Architecture for SafeScape 2.0 (adapted from Anohina-Naumeca, 2007).

SafeScape 2.0 implements four major components of an Intelligent Tutoring System (ITS) that work together to provide adaptive and interactive fire-safety learning. The Learner Model represents the learner's current knowledge state through the BKT knowledge vector, which is continuously updated based on observed responses. The Domain Model represents the fire-safety knowledge, competencies, scenarios, and associated learning content that constitute the subject matter of the system. The Tutoring and Adaptation Component uses the learner's estimated mastery to regulate learning activities through the DDA mechanism and provide appropriate feedback based on learner performance. Finally, the User Interface provides the interactive and gamified environment through which learners access scenarios, submit responses, receive feedback, and interact with the adaptive learning features. Collectively, these components establish the computational structure through which

SafeScape 2.0 models learner knowledge, represents domain competencies, adapts instruction, and delivers interactive learning experiences.

Bayesian Knowledge Tracing

Bayesian Knowledge Tracing (BKT) is a probabilistic learner-modeling technique used to estimate the likelihood that a learner has mastered a particular skill based on observed performance across successive learning opportunities. The model treats learner knowledge as a latent state, meaning that mastery cannot be observed directly but is inferred from the learner's responses to learning activities or assessment items. As additional response evidence becomes available, the model updates the estimated probability of mastery, allowing the learner's knowledge state to be continuously tracked throughout the learning process.

Building on the original knowledge-tracing framework of Corbett and Anderson, BKT maintains probabilistic estimates of a learner's mastery of individual skills and uses these estimates to support instructional decisions. Rather than interpreting a correct or incorrect response as definitive evidence of mastery or non-mastery, the model considers the uncertainty associated with learner performance. Thus, a correct response may provide evidence of learning without necessarily confirming mastery, while an incorrect response may not necessarily indicate that the learner has failed to acquire the skill.

In this study, BKT is adopted to estimate the learner's evolving mastery of specific skills based on responses to successive learning activities. The learner's knowledge at a given point in the learning process is represented by one of two latent states: Not Learned or Learned. Because the learner's actual knowledge state cannot be directly observed, the system infers the most probable state from the learner's

observed responses. Each new response serves as additional evidence that is incorporated into the learner's current knowledge estimate, resulting in an updated probability of mastery.

The use of BKT in the proposed system therefore provides a mechanism for tracking changes in learner knowledge over time rather than relying solely on raw scores or isolated responses. The resulting mastery estimates can serve as a basis for adaptive instructional decisions, such as determining whether the learner requires additional practice, remediation, or progression to subsequent learning activities. This makes BKT appropriate for the study's objective of providing learning experiences that respond to the learner's estimated level of skill acquisition.

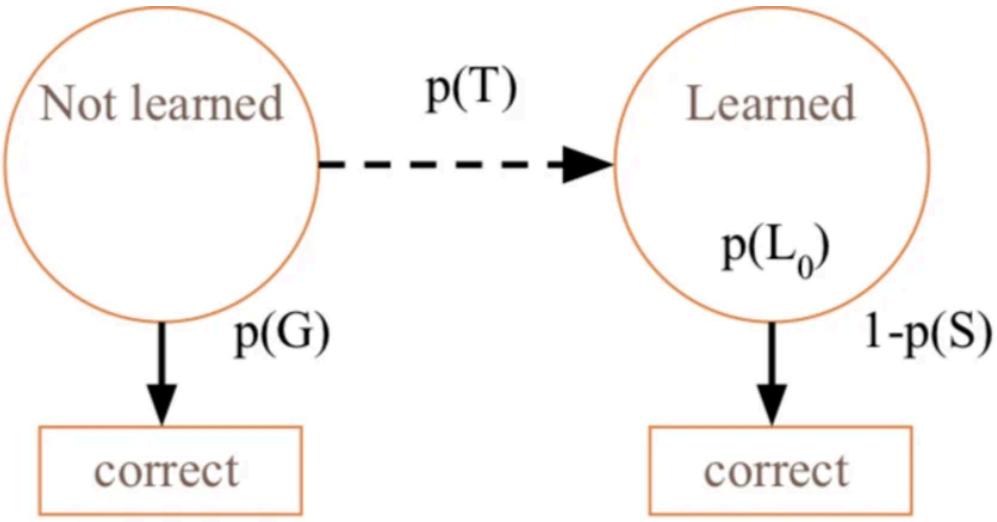


Figure X. Bayesian Knowledge Tracing State Model

https://www.researchgate.net/figure/Bayesian-Knowledge-Tracing-diagram-reproduced-from-Author-2015_fig1_329249607

As illustrated in Figure X, the BKT model consists of four principal parameters: the initial **knowledge probability p(L0)**, **learning probability p(T)**,

guess probability p(G), and **slip probability p(S)**. These parameters collectively describe the learner's initial knowledge, the likelihood of acquiring the skill, and the possibility of obtaining either a correct or incorrect response despite the learner's actual knowledge state.

The learner initially has a probability **(p(L_0))** of already knowing the target skill. If the learner is initially classified in the Not Learned state, there is a probability (p(T)) that the learner will transition to the Learned state as a result of instruction, practice, or interaction with the learning activity. The model assumes that once the learner has acquired the skill, the learner remains in the Learned state during the current tracing sequence.

The observed response is influenced by the learner's underlying knowledge state. When the learner is in the Not Learned state, a correct response may still occur with probability (p(G)), representing the learner guessing the correct answer despite not having mastered the skill. Conversely, the learner may provide an incorrect response with probability (1-p(G)). This distinction prevents the system from treating every correct answer as direct evidence of mastery.

When the learner is in the Learned state, the probability of producing a correct response is (1-p(S)), where (p(S)) represents the slip probability. A slip occurs when a learner who has already acquired the target skill nevertheless provides an incorrect response. Therefore, an incorrect response does not necessarily indicate that the learner has not learned the skill.

Table 3. Bayesian Knowledge Tracing Parameters Used for Learner Knowledge Estimation

Parameter	Description	Role in the Model
(p(L_0))	Initial knowledge probability	Represents the probability that the learner already possesses the target skill before completing the learning activity.
(p(T))	Learning or transition probability	Represents the probability that a learner in the Not Learned state acquires the target skill during the learning process.
(p(G))	Guess probability	Represents the probability that a learner in the Not Learned state answers an item correctly through guessing.
(p(S))	Slip probability	Represents the probability that a learner in the Learned state answers an item incorrectly despite possessing the skill.
(1-p(G))	Incorrect response probability for Not Learned	Represents the probability that a learner who has not learned the skill provides an incorrect response.
(1-p(S))	Correct response probability for Learned	Represents the probability that a learner who has learned the skill provides a correct response.

The model therefore distinguishes knowledge acquisition from observed performance. A correct response can originate from either a learner who has genuinely learned the skill or a learner who has not yet learned the skill but correctly guesses the answer. Similarly, an incorrect response can result either from a learner who has not learned the skill or from a learner who has already learned it but commits a slip.

This distinction is particularly important in the proposed adaptive learning system because learner performance is not determined solely by the number of correct or incorrect answers. Instead, the system uses the sequence of responses to estimate the learner's probability of mastery for the target skill. The resulting knowledge estimate can subsequently

be used to support decisions regarding the learner's succeeding activities, such as whether to proceed to a more advanced learning task, provide additional practice, or present remedial learning content.

In SafeScape 2.0, BKT is applied to estimate the learner's mastery of predefined fire-safety competencies from their responses during simulated learning activities. The resulting mastery estimates are continuously updated and used by the adaptive learning mechanism to inform the selection of subsequent scenario difficulty, thereby integrating learner modeling with personalized instructional delivery.

3.3.2 Knowledge-State Updating

The proposed system uses Bayesian Knowledge Tracing (BKT) to continuously update the learner's estimated mastery of a target skill after each learning interaction. Instead of determining mastery solely from the learner's raw score or the correctness of an individual response, the system maintains a probabilistic estimate of the learner's knowledge state and updates this estimate as new response evidence becomes available.

Let (P(L_{t-1})) denote the learner's estimated probability of having mastered the target skill after the previous interaction, where (t) represents the current learning interaction. Before incorporating the learner's current response, the model first accounts for the possibility that the learner acquired the skill during the current learning opportunity. The resulting prior probability of mastery is calculated as:

$$P(L_t^-) = P(L_{t-1}) + [1 - P(L_{t-1})]\,P(T)$$

where:

- *(P(L_{t-1})) is the learner's estimated probability of mastery after the previous interaction;*
- *(P(T)) is the probability that the learner acquires the target skill during the current learning opportunity; and*
- *(P(L_t^{-})) is the learner's estimated probability of mastery* ***before the current response is considered****.*

The term (1-P(L_{t-1})) represents the probability that the learner has not yet mastered the skill. Multiplying this value by (P(T)) estimates the probability that a previously unmastered learner acquires the skill during the current interaction. Thus, the equation combines the learner's existing mastery estimate with the possibility of newly acquired knowledge.

After the prior probability is obtained, the learner's observed response is incorporated into the model. The response update differs depending on whether the learner provides a correct or incorrect answer.

3.3.2.1 Correct Response

For a correct response, the posterior probability of mastery is calculated as:

$$P(L_t \mid Correct) = \frac{P(L_t^-)}{P(L_t^-) + \left[1 - P(L_t^-)\right] P(G)}$$

where (P(G)) represents the probability that a learner who has not yet mastered the skill answers the item correctly through guessing.

The equation reflects the fact that a correct response provides evidence of mastery but does not guarantee that the learner has mastered the skill. A learner in the Not Learned state may still answer correctly because of guessing. Therefore, the posterior probability of mastery depends not only on the observed correct response but also on the estimated probability of guessing.

3.3.2.2 Incorrect Response

For an incorrect response, the posterior probability of mastery is calculated as:v

$$P(L_t \mid Incorrect) = \frac{P(L_t^-)P(S)}{P(L_t^-)P(S) + \left[1 - P(L_t^-)\right]\left[1 - P(G)\right]}$$

where (P(S)) represents the probability that a learner who has already mastered the skill nevertheless provides an incorrect response because of a slip.

Consequently, an incorrect response does not automatically imply that the learner has failed to acquire the skill. The model considers the possibility that the learner knows the skill but made an error during the current interaction. This probabilistic treatment allows the system to avoid making immediate and potentially inaccurate conclusions about learner mastery based on a single response.

3.3.3 Sequential Knowledge-Update Process

The complete BKT updating process implemented in the proposed system can therefore be summarized as follows:

$$P(L_{t-1}) \rightarrow P(L_t^-) \rightarrow \text{Observed Response} \rightarrow P(L_t)$$

First, the learner's previous mastery estimate (P(L_{t-1})) is used to determine the prior mastery probability (P(L_t^{-})), incorporating the probability of learning (P(T)). Second, the learner's current response is evaluated. Third, the appropriate Bayesian update is applied using the guess probability (P(G)) for a correct response or the slip probability (P(S)) for an incorrect response. Finally, the resulting posterior probability (P(L_t)) becomes the learner's updated mastery estimate and serves as the starting point for the next learning interaction.

This sequential updating mechanism enables the proposed system to maintain an individualized and continuously updated representation of learner knowledge. As the learner completes additional activities, the system accumulates response evidence and progressively refines the estimated mastery probability for each target skill. The resulting estimate can then be used as an input to the system's adaptive-learning mechanism in determining whether the learner should continue practicing the current skill, receive additional or remedial activities, or proceed to a subsequent learning competency.

Table 4. Sequential Process for Updating Learner Knowledge Using Bayesian Knowledge Tracing

Step	Process	Output
1	Retrieve the learner's previous mastery probability (P(L_{t-1})).	Previous knowledge estimate
2	Apply the learning probability (P(T)).	Prior mastery probability (P(L_t^{-}))
3	Record the learner's current response.	Correct or Incorrect

Table 4. Continued….

Step	Process	Output
4	If the response is correct, apply (P(G)).	Updated mastery probability (P(L_t))
5	If the response is incorrect, apply (P(S)).	Updated mastery probability (P(L_t))
6	Store (P(L_t)) as the learner's current mastery estimate.	Updated learner knowledge state
7	Use the updated estimate for adaptive instructional decisions.	Next appropriate learning activity

3.3.4 Application of Bayesian Knowledge Tracing in the Proposed Adaptive Learning Platform

In the proposed adaptive learning platform, Bayesian Knowledge Tracing (BKT) is applied at the skill or competency level to continuously estimate the learner's mastery of specific learning targets. Each learning activity or assessment item is mapped to a corresponding target skill or competency. Before the learner begins a learning sequence, the system initializes the learner's estimated knowledge state using the initial knowledge probability, (p(L_0)). This value represents the learner's estimated probability of already possessing the target skill prior to the learning activities.

As the learner progresses through the learning sequence, the system records the response to each learning activity and updates the corresponding mastery probability using the BKT parameters. The updated probability reflects the accumulated evidence from the learner's previous and current responses. In this process, the model considers both the possibility that the learner has acquired the skill

and the possibility that the observed response does not perfectly represent the learner's actual knowledge state.

A correct response generally results in an increase in the estimated probability of mastery. However, the increase is influenced by the guess probability, (p(G)), because a learner who has not yet mastered the skill may still provide a correct answer through guessing. Therefore, a correct response is treated as evidence of learning rather than as definitive proof of mastery. Conversely, an incorrect response generally results in a reduction in the estimated mastery probability. However, this reduction is moderated by the slip probability, (p(S)), because a learner who has already mastered the skill may occasionally provide an incorrect response due to an error or slip.

Through this sequential updating process, BKT provides the system with a probabilistic estimate of learner mastery for each target skill rather than relying solely on the learner's percentage score or the result of an individual assessment item. The resulting mastery estimate provides a basis for the platform's adaptive-learning decisions. Depending on the estimated mastery level and the predefined decision criteria of the system, the learner may be directed to continue practicing the current competency, receive additional or remedial learning activities, or proceed to the next competency.

Thus, BKT serves as the learner-modeling component of the proposed platform by transforming successive learner responses into continuously updated knowledge estimates. These estimates enable the system to adjust the sequence, difficulty, and type of learning activities according to the learner's estimated knowledge state, supporting a more individualized and responsive learning experience.

The BKT output serves as an input to the Dynamic Difficulty Adjustment (DDA) component. Specifically, the estimated mastery probability generated by BKT provides the DDA component with evidence of the learner's current level of competency. The DDA component uses this information, together with the predefined difficulty criteria of the system, to determine the appropriate difficulty level of the learner's succeeding activity.

For example, a lower mastery probability may indicate that the learner requires additional practice, reinforcement, or a less difficult activity. A moderate mastery probability may result in the learner receiving activities at an appropriate or intermediate difficulty level to further develop the target skill. Conversely, a higher mastery probability may indicate sufficient evidence of mastery, allowing the DDA component to recommend a more challenging activity or progression to the next competency.

The integration can therefore be represented as:

$$\boxed{\text{Learner Response} \rightarrow \text{BKT} \rightarrow P(L_t) \rightarrow \text{DDA} \rightarrow \text{Next Activity Difficulty}}$$

Through this integration, BKT determines what the system estimates the learner knows, while DDA determines how the learning content should respond to that estimated knowledge. This separation allows the proposed platform to use the learner's continuously updated mastery estimate as a basis for dynamically selecting or adjusting subsequent learning activities. Consequently, the system can provide a more individualized learning sequence in which activity difficulty is responsive to the learner's evolving knowledge state.

3.4 Dynamic Difficulty Adjustment

The Dynamic Difficulty Adjustment (DDA) mechanism operationalizes the learner's continuous mastery probability estimated through Bayesian Knowledge Tracing into discrete scenario-difficulty levels. This approach is consistent with DDA research, which supports adjusting serious-game difficulty according to measurable indicators of learner or player capability and performance (Vahldick et al., 2017; Seyderhelm & Blackmore, 2024; Víteková et al., 2026).

In SafeScape 2.0, the mastery probability PL is mapped to Easy, Moderate, and Hard difficulty levels using predefined thresholds of 0.50 and 0.85. These thresholds are treated as system-specific operational parameters rather than universal BKT standards and will be evaluated empirically through functional and adaptive-effectiveness testing.

The implementation uses predefined mastery thresholds. A simplified representation is:

$$D(P_L) = \begin{cases} \text{Easy}, & 0 \leq P_L < 0.50, \\ \text{Moderate}, & 0.50 \leq P_L < 0.85, \\ \text{Hard}, & 0.85 \leq P_L \leq 1.00, \end{cases}$$

where:

- *D(PL) = difficulty level assigned by the Dynamic Difficulty Adjustment (DDA) mechanism;*
- *PL = learner's current mastery probability estimated by the Bayesian Knowledge Tracing (BKT) model;*
- ***Easy** = lower-complexity learning scenarios;*

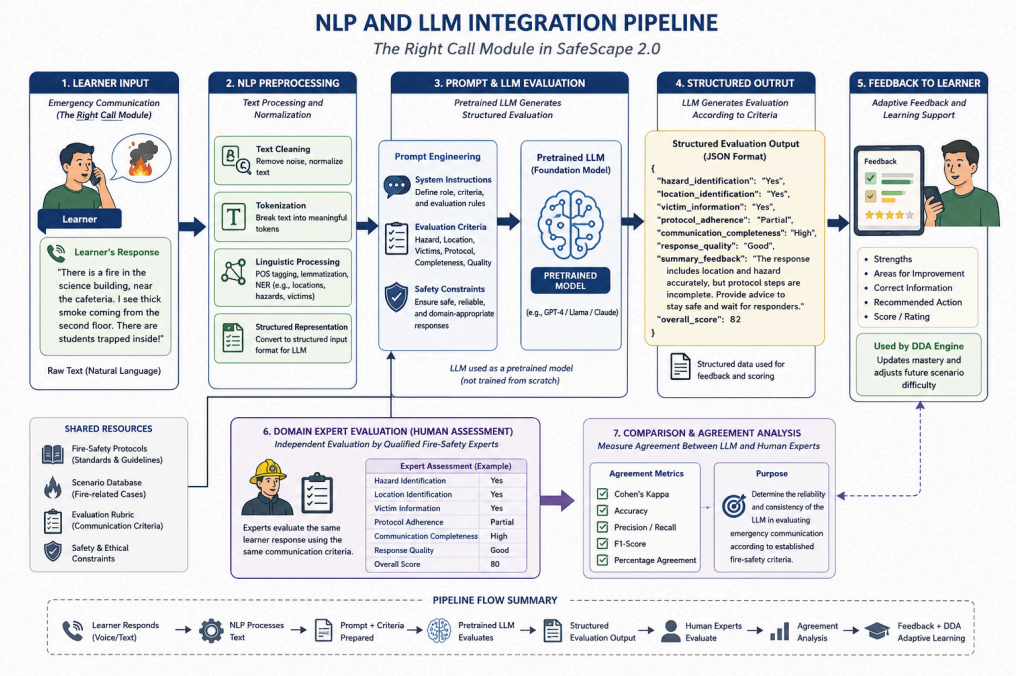
- ***Moderate*** *= intermediate-complexity learning scenarios; and*
- ***Hard*** *= higher-complexity learning scenarios.*

The DDA mechanism will dynamically adjust the characteristics of subsequent learning scenarios based on the learner's estimated mastery level. The adaptation may involve scenario complexity, number of distractors, time constraints, information density, hazard combinations, and caller complexity, allowing the system to progressively increase or decrease the cognitive and decision-making demands of the activity. The predefined mastery thresholds will serve as system-specific decision rules for mapping the continuous BKT mastery probability to discrete difficulty levels. These rules will be evaluated separately through functional testing, to verify that the intended difficulty level is correctly selected for each mastery condition, and effectiveness testing, to determine whether the resulting adaptation contributes to improved learner performance and learning outcomes.

3.5 Natural Language Processing and Large Language Models Integration

Natural Language Processing (NLP) and Large Language Model (LLM) techniques are integrated into the The Right Call module to simulate emergency communication and evaluate learner-generated responses within fire-related scenarios. The LLM receives the learner's emergency response and generates a structured evaluation based on predefined communication criteria, including hazard identification, location identification, victim or casualty information, adherence to established emergency protocols, completeness of the information provided, and overall response quality. The LLM is utilized as a pretrained model rather than being trained from scratch; therefore, the study focuses on its integration into the SafeScape 2.0 architecture, including prompt design, structured output generation, safety

constraints, and domain-specific validation. To establish the reliability of the automated evaluation, LLM-generated assessments will be compared with independent assessments from qualified fire-safety domain experts using appropriate agreement and classification measures. This validation process will determine the extent to which the LLM can consistently evaluate learner-generated emergency communication according to established fire-safety criteria.



Gamification and Simulation

SafeScape 2.0 incorporates game-based interaction as an instructional mechanism for delivering fire-safety concepts through interactive, scenario-based learning activities. Rather than presenting fire-safety procedures solely through conventional instructional materials, the system places learners in simulated emergency situations in which they are required to identify potential hazards, interpret situational information, select appropriate actions, and respond to evolving emergency

conditions. Each interaction produces observable learner responses that can be recorded and analyzed by the system.

The game-based environment is designed to support the repeated practice and application of fire-safety knowledge within controlled and progressively challenging scenarios. The system provides immediate feedback following learner decisions, allowing users to recognize errors, reinforce appropriate responses, and improve subsequent performance. Scenario difficulty may also be adjusted according to the learner's demonstrated performance, thereby supporting a progression from relatively simple safety-identification tasks to more complex emergency decision-making situations.

Specifically, the game-based component supports the following instructional functions:

- **Repeated practice** – provides multiple opportunities for learners to apply fire-safety concepts and procedures across different scenarios;
- **Immediate feedback** – communicates the correctness or appropriateness of learner responses following an interaction;
- **Progressive difficulty** – introduces increasingly complex scenarios or decision requirements as the learner demonstrates improved performance;
- **Scenario-based decision-making** – requires learners to apply acquired knowledge to contextualized fire-safety situations rather than relying solely on recall-based responses; and
- **Performance measurement** – records learner responses, decisions, and task outcomes as observable evidence of learning performance.

From a system perspective, game-based interaction serves as the learning activity and assessment environment, while Bayesian Knowledge Tracing (BKT) provides the mechanism for estimating the learner's evolving mastery of the targeted fire-safety competency. The learner's responses within the scenarios constitute observable evidence that is used to update the estimated knowledge state. The resulting BKT mastery estimate is then provided as an input to the Dynamic Difficulty Adjustment (DDA) component, which determines the appropriate level of difficulty and the subsequent learning activity. In this manner, the gamified environment is not itself the basis for determining mastery; rather, it provides the interaction context through which learner performance is generated, observed, and subsequently processed by the adaptive components.

Accordingly, gamification is treated as a supporting instructional mechanism rather than the primary adaptive mechanism of SafeScape 2.0. Its principal function is to facilitate learner engagement, repeated interaction, contextualized practice, and collection of performance evidence. The evaluation of the adaptive component therefore focuses on the operation of BKT for learner knowledge estimation and DDA for difficulty and activity selection, while the game-based environment provides the instructional context in which these mechanisms operate. This separation establishes a clearer distinction between the instructional interface, the learner-modeling mechanism, and the adaptive decision mechanism of the proposed system.

Gamification is therefore treated as a supporting instructional mechanism rather than the primary independent variable when evaluating BKT and DDA.

3.6 Data Persistence

The proposed system implements JSON-based local data persistence to maintain essential learner-related information throughout the learning process and preserve the learner's adaptive state across sessions. The persistence mechanism stores the learner's progression, total score, timestamp, and BKT knowledge vector. Learner progression represents the learner's current status across the available learning activities or competencies, while the total score records the cumulative performance achieved during completed activities. The timestamp identifies when the learner state was saved, and the BKT knowledge vector contains the estimated mastery probabilities associated with the target skills or competencies generated by the Bayesian Knowledge Tracing (BKT) component.

Among these stored variables, the BKT knowledge vector is particularly important to the adaptive mechanism because it preserves the learner's estimated knowledge state between sessions. When the application is reopened, the system retrieves the previously stored JSON data and restores the learner's progression, performance information, and corresponding knowledge estimates, allowing the adaptive learning process to continue from the learner's previously established state rather than treating the learner as a new user.

Thus, local data persistence supports session continuity and state preservation, enabling the BKT component to continue updating the learner's knowledge estimates and providing the Dynamic Difficulty Adjustment (DDA) component with the learner state required for subsequent adaptive decisions.

3.7 Algorithm Analysis

Algorithm analysis will be conducted to systematically evaluate the computational procedures implemented in SafeScape 2.0 and to determine whether they satisfy the fundamental characteristics of a valid algorithm while demonstrating appropriate computational efficiency under the intended operating conditions. The analysis will focus on the principal algorithms supporting the system's adaptive learning functions, particularly the Bayesian Knowledge Tracing (BKT) knowledge-state updating and the Dynamic Difficulty Adjustment (DDA) procedures. Each algorithm will be examined based on its defined inputs, processing steps, outputs, termination conditions, and computational requirements. This analysis will establish whether the implemented procedures are logically defined, computationally feasible, and capable of producing consistent and expected results for valid learner inputs.

Specifically, each algorithm will be evaluated according to the following criteria: (1) Input, referring to the data and parameters required for algorithm execution; (2) Output, referring to the expected result generated from the supplied inputs; (3) Definiteness, referring to whether each computational operation and decision rule is precisely specified and unambiguous; (4) Finiteness, referring to whether the algorithm terminates after a finite sequence of computational steps; (5) Effectiveness, referring to whether each operation is sufficiently defined and contributes to the intended computational objective; (6) Correctness, referring to the algorithm's ability to produce the expected output for valid inputs and defined conditions; (7) Time Complexity, referring to the computational effort or number of operations required as the size of the input increases; (8) Space Complexity, referring

to the amount of memory required for data storage and intermediate computations during execution; and (9) Performance Metrics, referring to empirically observable measures such as execution or response time and resource utilization during actual system operation.

The results of the algorithm analysis will provide both a theoretical and empirical basis for evaluating the computational suitability of the implemented procedures. The theoretical assessment will examine the logical properties, correctness, and asymptotic computational complexity of the algorithms, while the empirical assessment will consider their observed execution behavior within the SafeScape 2.0 environment. This approach is particularly relevant to the proposed adaptive architecture because the BKT and DDA procedures are repeatedly executed as learners interact with learning activities. Therefore, their computational efficiency and correctness may directly influence the system's responsiveness and its ability to provide timely, personalized learning activities.

3.7.1 Bayesian Knowledge Tracing (BKT) Algorithm

The Bayesian Knowledge Tracing (BKT) algorithm is implemented to estimate and continuously update the learner's probability of mastery of a specific target skill based on observed performance during learning activities. The algorithm receives as primary inputs the learner's previous mastery probability (P(L_{t-1})), the observed response correctness, and the predefined BKT parameters, namely the learning probability (P(T)), guess probability (P(G)), and slip probability (P(S)). These inputs represent the learner's current estimated knowledge state and the probabilistic factors that account for learning, guessing, and response errors.

During execution, the algorithm first incorporates the probability that learning occurs during the current interaction to obtain the learner's prior mastery probability (P(L_t^{-})). It then incorporates the observed response using Bayesian updating. For a correct response, the algorithm considers the possibility that the learner answered correctly through guessing; for an incorrect response, it considers the possibility that the learner possesses the skill but committed a slip. The resulting posterior probability, (P(L_t)), represents the learner's updated probability of mastering the target skill.

The updated mastery probability is subsequently returned by the algorithm and stored as the learner's current knowledge state. This value serves as the input to the next BKT update and may also be provided to the **Dynamic Difficulty Adjustment (DDA)** component for subsequent instructional decision-making. Thus, the BKT algorithm establishes a sequential learner-modeling process in which each observed response contributes probabilistic evidence to the learner's evolving knowledge state.

Algorithm 1. Bayesian Knowledge Tracing Knowledge-State Update

The Bayesian Knowledge Tracing (BKT) Knowledge-State Update algorithm is designed to estimate and update the learner's probability of mastery of a specific target competency based on the learner's previously estimated knowledge state and the observed response to a learning activity. The algorithm incorporates the learner's current mastery probability together with the predefined learning, guessing, and slip probabilities to determine an updated posterior mastery probability. This updated value represents the learner's estimated knowledge state after the current interaction and serves as the basis for subsequent BKT updates. In the proposed system, the resulting mastery estimate is also provided to the **Dynamic Difficulty Adjustment**

(DDA) component to support the selection of an appropriate subsequent learning activity or difficulty level.

Table 5. *Bayesian Knowledge Tracing Knowledge-State Update*

Element	Description
Input	Current mastery probability (P(L_{t-1})), response correctness (R_t), learning probability (P(T)), guess probability (P(G)), and slip probability (P(S))
Output	Updated mastery probability (P(L_t))
Response Values	(R_t = 1) for Correct; (R_t = 0) for Incorrect
Time Complexity	(O(1)) per skill update; (O(K)) for sequential updates of (K) competencies
Space Complexity	(O(K)) for maintaining the knowledge vector

Algorithm 1: BKT Knowledge-State Update

```
Algorithm 1: BKT Knowledge-State Update


Input:
   P(L_{t-1}) ← current mastery probability
   R_t        ← learner response correctness
   P(T)       ← learning probability
   P(G)       ← guess probability
   P(S)       ← slip probability


Output:
   P(L_t)     ← updated mastery probability
```

Begin

1. Compute the probability of mastery before observing the response:

$$P(L_t^-) \leftarrow P(L_{t-1}) + [1 - P(L_{t-1})] \times P(T)$$

2. If R_t = Correct then

Compute:

$$P(L_t) \leftarrow P(L_t^-) / [P(L_t^-) + (1 - P(L_t^-)) \times P(G)]$$

Else if R_t = Incorrect then

Compute:

$$P(L_t) \leftarrow [P(L_t^-) \times P(S)] / [P(L_t^-) \times P(S) + (1 - P(L_t^-)) \times (1 - P(G))]$$

End If

3. Store $P(L_t)$ as the learner's updated mastery probability.

4. Return $P(L_t)$.

End

Operational Flow of the Algorithm

The BKT procedure begins by retrieving the learner's previous mastery probability, (P(L_{t-1})), for the competency associated with the current learning activity. Before incorporating the learner's response, the algorithm calculates the pre-response mastery probability, (P(L_t^-)), by accounting for the possibility that the learner acquired the competency during the current learning opportunity. This is calculated using the learning probability (P(T)).

The algorithm then evaluates the learner's observed response. If the response is correct, the posterior mastery probability is calculated while accounting for the possibility that the learner arrived at the correct answer through guessing, represented by (P(G)). If the response is incorrect, the algorithm considers the possibility that the learner actually possesses the competency but made an error, represented by the slip probability (P(S)). The resulting posterior probability (P(L_t)) becomes the learner's updated knowledge state.

The updated value is subsequently stored in the learner's BKT knowledge vector and becomes (P(L_{t-1})) for the succeeding interaction involving the same competency. In the proposed adaptive learning architecture, this updated mastery estimate can then be passed to the Dynamic Difficulty Adjustment (DDA) component as an input for determining the appropriate difficulty level or subsequent learning activity.

Computational Complexity

For a single competency, the BKT update consists of a fixed sequence of arithmetic operations and conditional evaluation. Therefore, the time complexity of updating one competency is:

$$T(K = 1) = O(1)$$

where the computational cost does not increase with the number of learner responses for an individual update.

If the system maintains and updates (K) competencies sequentially, the total computational complexity is approximately:

[
T(K)=O(K)
]

where (K) represents the number of tracked competencies.

The BKT knowledge vector maintains one mastery probability for each competency. Consequently, the memory requirement is:

[
S(K)=O(K)
]

Thus, the algorithm has **constant time and space requirements for an individual competency update**, while the overall computational and memory

requirements scale linearly with the number of competencies tracked by the system. This computational characteristic makes the BKT procedure suitable for repeated execution during learner-system interactions in an adaptive learning environment.

3.8 DDA Algorithm

The **Dynamic Difficulty Adjustment (DDA) algorithm** is designed to determine the appropriate difficulty level of the next learning activity based on the learner's current mastery probability estimated by the **Bayesian Knowledge Tracing (BKT)** component. The algorithm receives the learner's current BKT mastery probability as input and compares this value against predefined mastery thresholds corresponding to the difficulty levels established in the system. Based on the applicable decision rule, the algorithm assigns the appropriate difficulty level and returns it as the output. In this manner, the DDA component translates the learner's estimated knowledge state into an actionable instructional decision, allowing the system to provide learning activities that are aligned with the learner's current level of competency.

Algorithm 2: Dynamic Difficulty Adjustment

Input:

P(L_t) ← learner's current BKT mastery probability

T1, T2, ..., Tn ← predefined mastery thresholds

D1, D2, ..., Dn+1 ← predefined difficulty levels

Output:

D ← selected difficulty level

```
Begin

1. Receive the current mastery probability P(L_t).

2. Compare P(L_t) with the predefined mastery thresholds.

3. If P(L_t) satisfies the condition for D1 then
      D ← D1

   Else if P(L_t) satisfies the condition for D2 then
      D ← D2

   ...

   Else if P(L_t) satisfies the condition for Dn then
      D ← Dn

   Else
      D ← Dn+1

4. Return D as the selected difficulty level.

End
```

The DDA algorithm receives the learner's current mastery probability, P(Lt), from the BKT component and maps this value to one of the predefined difficulty levels according to the system's established decision rules. The mastery probability serves as the basis for determining the learner's current estimated level of competency. The algorithm sequentially

evaluates the mastery value against the predefined thresholds until the corresponding difficulty condition is satisfied. The resulting difficulty classification is then returned to the adaptive learning module and used to select an appropriate subsequent learning activity.

To ensure deterministic behavior, the predefined mastery thresholds must collectively cover the valid range of the BKT mastery probability. Consequently, every valid mastery value must correspond to exactly one difficulty level, preventing overlapping or undefined decision conditions. The algorithm must also terminate after a finite number of conditional evaluations and return a difficulty level that is consistent with the predefined decision rules.

Because the DDA procedure evaluates a fixed number of conditional comparisons, its computational cost for a single decision is constant. Thus, its time complexity is:

O(1)

The algorithm requires only a constant amount of auxiliary memory for the mastery value, threshold comparisons, and selected difficulty level. Therefore, its auxiliary space complexity is:

O(1)

In the proposed system architecture, the overall adaptive sequence can therefore be represented as:

$$\boxed{\text{Learner Response} \rightarrow \text{BKT Update} \rightarrow P(L_t) \rightarrow \text{DDA} \rightarrow \text{Difficulty Level} \rightarrow \text{Next Activity}}$$

3.8.1 Large Language Model (LLM) and Natural Language Processing (NLP) Process

The Large Language Model (LLM) and Natural Language Processing (NLP) component is evaluated separately from the Bayesian Knowledge Tracing (BKT) and

Dynamic Difficulty Adjustment (DDA) algorithms because its primary language-processing capability is provided by a pretrained external language model. Consequently, the computational complexity of the underlying LLM architecture is not considered a directly controllable property of the proposed system. Instead, the evaluation focuses on the functional performance, reliability, responsiveness, and output quality of the LLM/NLP component as integrated into the proposed system.

The evaluation will assess the LLM/NLP component using measurable performance indicators, including response latency, successful API response rate, structured-output validity, classification accuracy, precision, recall, F1-score, and agreement with expert assessment. Response latency will measure the time required by the system to receive and process an LLM response, while successful API response rate will determine the proportion of requests that are completed successfully without API or processing errors. Structured-output validity will assess whether the generated response conforms to the required format, fields, and data structure defined by the system. These measures are particularly important because the LLM output may serve as an input to subsequent system processes and therefore must be both computationally accessible and structurally valid.

For tasks involving categorical classification, the predicted outputs generated by the LLM/NLP component will be compared with the corresponding ground-truth labels or expert-established classifications. The evaluation will use a confusion-matrix-based analysis consisting of true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN). From these values, precision, recall, and F1-score will be calculated.

Precision measures the proportion of instances classified as positive by the system that are actually positive:

$$Precision = \frac{TP}{TP + FP}$$

Recall measures the proportion of actual positive instances that are correctly identified by the system:

$$Recall = \frac{TP}{TP + FN}$$

The F1-score provides a harmonic mean of precision and recall and is used to provide a balanced measure when both types of classification errors are relevant:

$$F1 = 2\left(\frac{Precision \times Recall}{Precision + Recall}\right)$$

where ***TP*** represents true positives, FP represents false positives, and FN represents false negatives.

In addition to these quantitative measures, the LLM/NLP component will be evaluated based on its agreement with expert assessment. Expert assessment will serve as a reference for determining whether the system-generated classification or interpretation is consistent with the expected instructional or domain-specific judgment. This evaluation provides a complementary measure of output quality because numerical classification metrics alone may not fully capture the appropriateness or contextual relevance of generated language-based outputs.

The LLM/NLP evaluation is intended to determine whether the component can reliably process learner-related inputs, produce valid structured outputs, generate accurate classifications, and provide responses consistent with expert judgment. Unlike the BKT and DDA components, whose computational procedures and complexity are directly defined within the proposed system, the LLM/NLP component is therefore evaluated primarily according to its observable operational and predictive performance within the system's intended use context.

3.9 Data Collection Methods

Data collection will be conducted in several stages to support both system development and empirical evaluation.

3.9.1 Data Sources

The study will utilize multiple complementary data sources to support the development, implementation, evaluation, and validation of SafeScape 2.0. These sources will provide the empirical, domain-specific, learner-related, and system-level data necessary to assess the functionality and effectiveness of the proposed adaptive learning platform.

Table 6. Data Sources and Their Intended Use in the Study

No.	Data Source	Description	Purpose / Intended Use

1	Fire-Safety and Emergency-Response Materials	Instructional materials, guidelines, and reference documents containing established fire-safety concepts and emergency-response procedures.	To serve as the content and instructional basis for the learning activities and scenarios implemented in SafeScape 2.0.
2	Validated Domain Knowledge from Qualified BFP Personnel	Expert-validated information provided by qualified Bureau of Fire Protection (BFP) personnel.	To validate the accuracy, relevance, and appropriateness of fire-safety concepts, procedures, and emergency scenarios.
3	Fire-Safety Scenarios Developed for SafeScape 2.0	Structured scenario-based activities representing relevant fire-safety and emergency-response situations.	To provide learners with contextualized opportunities for knowledge application, decision-making, and repeated practice.
4	Learner Pre-Test and Post-Test Responses	Learner assessment results collected before and after the learning intervention.	To determine changes in learner performance and learning outcomes following interaction with the system.
5	Learner Interaction Logs	System-generated records of learner responses, activities, interactions, and progression.	To analyze learner behavior, interaction patterns, and progression throughout the learning process.
6	BKT Knowledge-State Records	Estimated learner mastery probabilities generated by the Bayesian Knowledge Tracing (BKT) component.	To monitor the evolution of the learner's estimated knowledge state across targeted competencies.
7	DDA Difficulty-Transition Records	Records of difficulty-level selections and transitions generated by the Dynamic Difficulty Adjustment (DDA) component.	To examine how the system adapts learning difficulty according to the learner's estimated mastery.
8	LLM-Generated Evaluations	Outputs generated by the integrated Large Language Model (LLM)/Natural Language Processing (NLP) component.	To evaluate the language-based assessment or classification functions implemented in SafeScape 2.0.

Table 6. Continued…

No.	Data Source	Description	Purpose / Intended Use

9	Expert Evaluations of LLM Outputs	Evaluations of LLM-generated outputs performed by qualified experts using predefined criteria or reference classifications.	To establish a reference basis for assessing the accuracy, appropriateness, and consistency of LLM-generated outputs.
10	Usability and User-Acceptance Responses	Responses obtained from intended users regarding their experience with the proposed system.	To assess the system's usability, acceptability, functionality, and perceived user experience.
11	Software Performance and Testing Logs	Records generated during software testing and system execution, including relevant performance observations and errors.	To evaluate the technical performance, reliability, responsiveness, and operational behavior of the system.

The study outlined in the provided documents employs multiple complementary data sources to support the development, implementation, evaluation, and validation of an adaptive learning platform named SafeScape 2.0. These sources are intended to gather empirical, domain-specific, learner-related, and system-level data to thoroughly assess the platform's functionality and effectiveness.

To establish the foundational content, the study utilizes established fire-safety and emergency-response materials as the instructional basis for the learning activities. The accuracy, relevance, and appropriateness of these fire-safety concepts and procedures are ensured through domain knowledge validated by qualified Bureau of Fire Protection (BFP) personnel. This validated knowledge is then integrated into structured fire-safety scenarios developed specifically for SafeScape 2.0, which provide learners with contextualized opportunities for knowledge application, decision-making, and repeated practice.

To assess the impact of the system on users, the study collects pre-test and post-test responses to determine changes in learner performance and learning outcomes following the intervention. This is supported by system-generated learner interaction logs used to analyze user behavior, interaction patterns, and progression throughout the learning process. Additionally, the platform uses Bayesian Knowledge

Tracing (BKT) to generate knowledge-state records that monitor the evolution of a learner's estimated mastery across targeted competencies. This mastery estimation directly informs the Dynamic Difficulty Adjustment (DDA) component, and the study collects difficulty-transition records to examine how the system actively adapts the learning difficulty for the user.

Finally, the study incorporates comprehensive data sources to evaluate the system's technical components and overall usability. It collects outputs from the integrated Large Language Model (LLM) and Natural Language Processing (NLP) components to evaluate the system's language-based assessment functions. To verify these functions, qualified experts evaluate the LLM-generated outputs using predefined criteria, establishing a reference basis for accuracy and consistency. The overall acceptance of the proposed system is assessed through usability and user-acceptance responses gathered from intended users. Alongside this user feedback, software performance and testing logs are generated during execution to evaluate the technical performance, responsiveness, reliability, and operational behavior of the system.

Table 7. Data Categories

Dataset	Attribute	Description	Values/Format
Learner Profile	Participant ID	Anonymous identifier	Alphanumeric
Learner Profile	Group	Experimental/comparison	Categorical
Learner Profile	Session	Gameplay session	Integer

Table 7. Continued…

Dataset	Attribute	Description	Values/Format
Knowledge Test	Pre-test Score	Baseline safety knowledge	Numerical
Knowledge Test	Post-test Score	Post-intervention knowledge	Numerical
BKT	Skill ID	Fire-safety competency	Categorical
BKT	Mastery Probability	Estimated learner mastery	0–1
BKT	Response	Learner response outcome	Correct/Incorrect
DDA	Difficulty Level	Current scenario difficulty	Easy/Moderate/Hard
DDA	Transition	Difficulty change	Up/Down/Same
Gameplay	Response Time	Time to complete task	Seconds
Gameplay	Score	Scenario/game score	Numerical
LLM	User Response	Emergency communication response	Text
LLM	Hazard Classification	Identified hazard	Categorical
LLM	Protocol Score	Protocol adherence	Numerical
LLM	Communication Score	Communication quality	Numerical
Expert	Expert Rating	Human assessment of LLM output	Numerical/Categorical
Usability	SUS Score	System usability	0–100
UAT	Acceptance Rating	User assessment	1–5
Performance	API Latency	Response time	Milliseconds
Performance	FPS	Rendering performance	Frames/sec

The Data Categories table outlines the structured datasets collected to evaluate SafeScape 2.0 across its pedagogical, artificial intelligence, and technical dimensions. The Learner Profile dataset utilizes anonymous alphanumeric IDs to ensure data

privacy while tracking gameplay sessions. Within this profile, the Group attribute is critical for the quasi-experimental design, allowing researchers to separate participants into experimental (adaptive) and comparison (non-adaptive) conditions. To measure learning outcomes, the Knowledge Test dataset captures pre-test and post-test scores, providing the foundational metrics needed to assess baseline fire safety knowledge and subsequent learning gains.

During gameplay, the Bayesian Knowledge Tracing (BKT) attributes track the continuous evolution of a learner's cognitive state. By recording the specific fire-safety competency (Skill ID) and the correctness of the learner's response, the system generates a continuously updated Mastery Probability bounded between 0 and 1. This probability serves as the direct input for the Dynamic Difficulty Adjustment (DDA) dataset, which logs the selected difficulty level—Easy, Moderate, or Hard—and tracks procedural transitions (Up, Down, or Same) to ensure the scenario complexity scales appropriately with the user's demonstrated skill.

Furthermore, the LLM dataset captures the raw, natural language emergency communications provided by users in Taglish. It records the AI's structured evaluation of these inputs, including categorical Hazard Classifications and numerical scores for Protocol Adherence and Communication Quality. To establish the validity of these AI-generated evaluations, the Expert dataset logs independent human assessments from BFP personnel, allowing for statistical agreement analysis between the system and domain experts. Finally, the evaluation of user experience and system quality relies on the Usability and UAT datasets, which record quantitative System Usability Scale (SUS) scores and user acceptance ratings. The Performance dataset monitors the technical stability of the hybrid architecture by logging rendering performance (FPS)

and API Latency, ensuring that the integration of external machine learning microservices does not introduce bottlenecks or thread blockages on low-specification devices.

3.9.2 Expert Data Collection

The study will employ expert data collection to establish the domain validity and contextual appropriateness of the fire-safety and emergency-response components incorporated into SafeScape 2.0. Qualified fire-safety personnel will be engaged as domain experts to review and evaluate selected system components based on predefined evaluation criteria. Their assessment will focus on the accuracy and appropriateness of fire-safety content, realism and contextual relevance of emergency scenarios, conformity of emergency-response procedures with established protocols, appropriateness of hazard classifications, clarity and adequacy of communication criteria, and accuracy and appropriateness of outputs generated by the integrated LLM/NLP component.

The expert-generated data will be documented and analyzed separately from learner-generated data. This separation is methodologically important because expert evaluation serves as evidence of domain validity and content appropriateness, whereas learner data are used to evaluate learning performance, knowledge acquisition, adaptive behavior, usability, and educational effectiveness. Maintaining separate datasets therefore prevents the conflation of domain-validation measures with learner-outcome measures and provides a clearer basis for interpreting the results of the study.

Where applicable, expert assessments will be summarized using appropriate descriptive and agreement measures to determine the extent to which the system's

content, scenarios, protocols, classifications, communication criteria, and LLM/NLP outputs conform to established fire-safety knowledge and expert judgment.

3.9.3 Learner Data Collection

Learner data will be collected during the controlled evaluation of SafeScape 2.0 to obtain empirical evidence regarding learner performance, knowledge development, and the operation of the system's adaptive mechanisms. Prior to the intervention, participants will complete the prescribed pre-test, which will establish their baseline performance on the targeted fire-safety competencies. Following completion of the prescribed gameplay and learning activities, participants will undertake the post-test to provide a basis for determining changes in performance after exposure to the system.

Throughout the gameplay sessions, SafeScape 2.0 will automatically capture and record relevant learner interaction data. These data will include learner responses, activity scores, response times, BKT-derived mastery probabilities or knowledge states, difficulty-level transitions generated by the DDA component, scenario outcomes, and other relevant interaction events. Automated logging is intended to minimize reliance on manual recording and to provide a consistent record of learner-system interactions.

The resulting learner dataset will be used to analyze learner performance, changes in knowledge, progression across competencies, and the behavior of the adaptive learning mechanism. In particular, the BKT knowledge-state records will provide evidence of the learner's estimated mastery over successive interactions, while the DDA transition records will document how changes in estimated mastery are translated into subsequent difficulty-level decisions. The interaction logs will

therefore complement the pre-test and post-test results by providing process-level evidence of how learners interacted with SafeScape 2.0 during the intervention, rather than relying solely on their aggregate assessment scores.

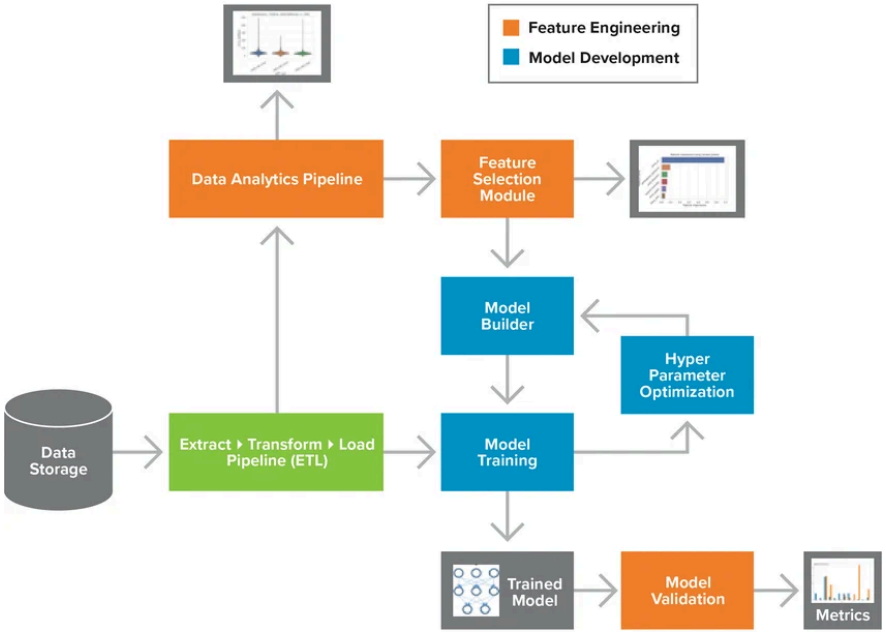
3.9.4 Data Privacy

Participant information will be represented using anonymous identifiers rather than names in the research dataset. Personally identifiable information will not be included in analytical datasets unless required for approved research administration.

For LLM-based processing, only the information necessary for the evaluation task will be transmitted. Where applicable, identifying information will be removed or anonymized before processing. Research data will be stored securely and accessed only by authorized researchers.

3.10 Data Model Generation

The study will implement a **structured artificial intelligence and machine learning (AI/ML) development and evaluation workflow** to guide the systematic development, integration, and assessment of the computational components of SafeScape 2.0. The workflow will encompass **data collection, data preparation, model and parameter configuration, algorithm implementation, component-level evaluation, system integration, and deployment**. This approach ensures that each computational component is developed and evaluated according to its intended function within the proposed adaptive learning platform.



https://www.researchgate.net/figure/General-workflow-for-developing-the-ML-model-for-an-AI-ML-solution_fig3_370833924

SafeScape 2.0 integrates three computational approaches, namely Bayesian Knowledge Tracing (BKT), Dynamic Difficulty Adjustment (DDA), and Large Language Model/Natural Language Processing (LLM/NLP). These approaches differ in their computational mechanisms and therefore require distinct development and evaluation procedures. BKT is implemented as a probabilistic learner-modeling mechanism that estimates the learner's evolving mastery of targeted competencies from observed responses. DDA is implemented as a deterministic rule-based mechanism that uses the BKT-derived mastery probability to determine the appropriate difficulty level of subsequent learning activities. Meanwhile, the LLM/NLP component utilizes a pretrained external language model to perform designated language-processing tasks and is evaluated based on predefined performance and validity criteria.

Because of these differences, the study will not conceptualize BKT, DDA, and LLM/NLP as a single model-training pipeline. Instead, each component will be subjected to a methodological procedure appropriate to its underlying computational approach. The BKT component will undergo parameter configuration and knowledge-state evaluation; the DDA component will undergo verification of its decision rules, threshold conditions, and adaptive transitions; and the LLM/NLP component will undergo structured evaluation of its generated outputs using quantitative performance measures and expert assessment.

The interaction among these components forms the core of the adaptive mechanism of SafeScape 2.0. Learner responses serve as evidence for updating the BKT knowledge state; the resulting mastery probability is then provided to the DDA component to determine the appropriate difficulty of the next learning activity. Where applicable, the LLM/NLP component processes designated language-based inputs and generates structured outputs that are evaluated for accuracy, validity, and agreement with expert judgment. Thus, the workflow emphasizes not only the individual performance of each computational component but also their functional integration within the overall adaptive learning architecture.

The resulting workflow provides a systematic basis for evaluating SafeScape 2.0 at both the component level and system level. Component-level evaluation determines whether BKT, DDA, and LLM/NLP perform according to their respective specifications, while system-level evaluation examines whether their integration supports the intended adaptive learning process. This methodological separation strengthens the validity of the evaluation by ensuring that probabilistic modeling,

rule-based adaptation, and pretrained language-model processing are assessed using criteria appropriate to their respective computational characteristics.

3.10.1 Data Preparation

The data preparation stage will involve the **systematic organization, structuring, and representation of fire-safety information** required for the development and implementation of SafeScape 2.0. Collected information from relevant fire-safety and emergency-response materials, together with domain-validated knowledge, will be reviewed and transformed into a structured knowledge representation suitable for integration into the system's learning content, adaptive mechanisms, and evaluation processes.

The prepared knowledge representation will include the following elements: **competency, skill, scenario, hazard, expected response, difficulty level, feedback, prerequisite relationship, and evaluation criteria**. A **competency** will represent a broader fire-safety learning objective, while a **skill** will represent a specific knowledge or performance component associated with that competency. **Scenarios** will operationalize these competencies through contextualized fire-safety situations, while **hazards** will identify the relevant risks or unsafe conditions presented within each scenario. The **expected response** will define the appropriate learner action or decision based on the established fire-safety knowledge.

Each learning scenario will also be associated with a **predefined difficulty level** to support the Dynamic Difficulty Adjustment (DDA) mechanism. Corresponding **feedback** will be prepared to provide learners with information regarding the appropriateness of their responses. **Prerequisite relationships** will be

established where a particular skill or competency requires prior knowledge or mastery of another skill. Finally, **evaluation criteria** will define the basis for determining whether a learner response or system-generated output satisfies the expected requirements.

Each scenario will be systematically **mapped to one or more target fire-safety competencies and their corresponding skills**. This mapping will establish the relationship between instructional content and the learner model, allowing learner responses to specific activities to be associated with the appropriate competency tracked by the BKT component. The resulting structured representation will therefore serve as a common basis for **content delivery, learner-state estimation, adaptive difficulty selection, feedback generation, and system evaluation**.

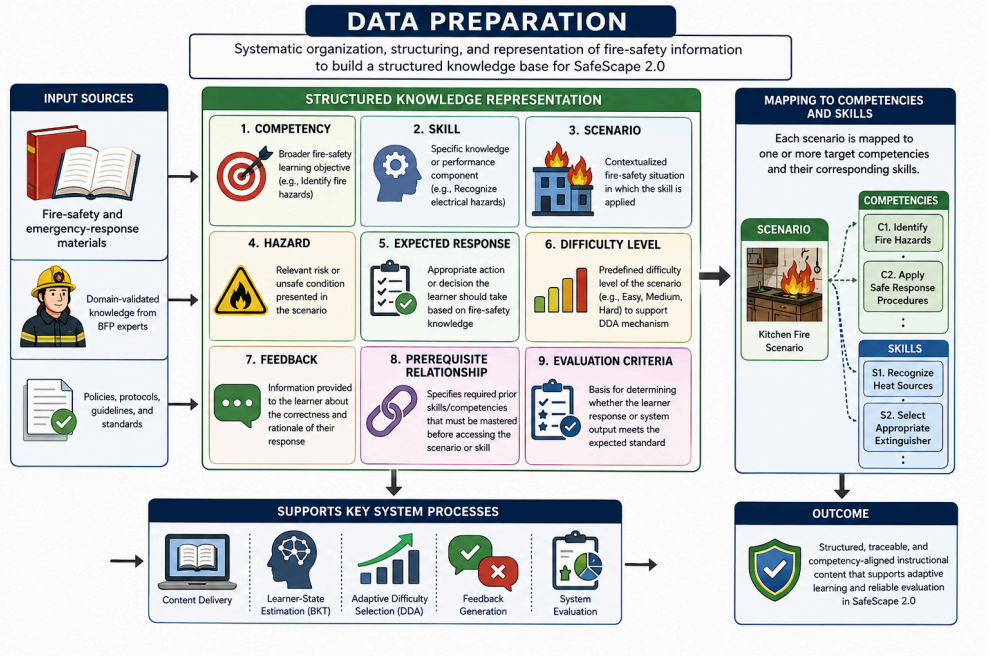
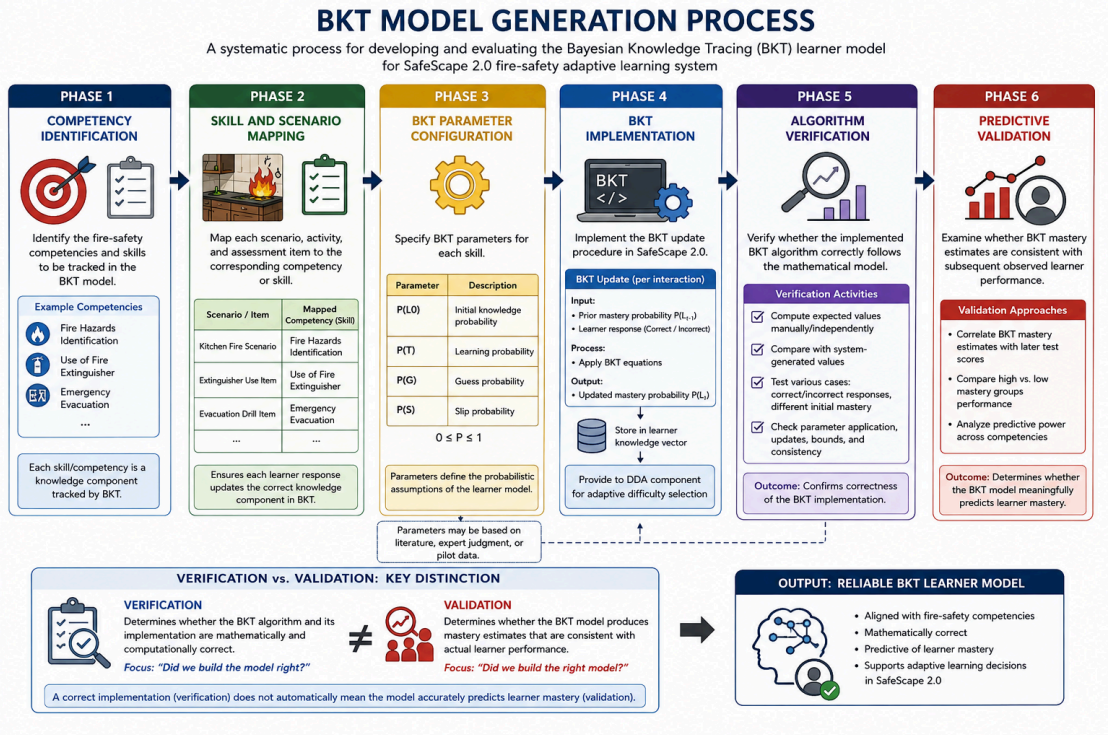


Figure x. Data Preparation Pipeline

This preparation process will ensure that the instructional content used by SafeScape 2.0 is **structured, traceable, and consistently aligned with the targeted learning competencies and adaptive-learning requirements**.

3.10.2 BKT Model Generation

The Bayesian Knowledge Tracing (BKT) model generation process will be conducted through a systematic sequence of competency identification, skill mapping, parameter configuration, implementation, verification, and predictive validation. This process is intended to ensure that the learner model is appropriately aligned with the fire-safety competencies targeted by SafeScape 2.0 and that its computational implementation produces reliable and interpretable knowledge-state estimates.



Phase 1 – Competency Identification.

The study will first identify the specific fire-safety competencies and skills that will be represented in the learner model. Only competencies that are explicitly addressed

by the learning activities and assessment items will be included in the BKT model. Each tracked skill will represent a distinct knowledge component for which the learner's mastery can be estimated over successive learning interactions.

Phase 2 – Skill and Scenario Mapping.

The identified skills will be mapped to the corresponding learning scenarios, activities, and assessment items. This mapping will establish the relationship between each learner response and the specific competency or skill being assessed. It will also ensure that the knowledge state updated by BKT corresponds to the intended instructional target rather than to an unrelated activity or scenario.

Phase 3 – BKT Parameter Configuration.

The BKT parameters will then be specified for each tracked skill, including the initial knowledge probability, learning probability, guess probability, and slip probability. These parameters will define the probabilistic assumptions governing the learner's initial knowledge state, the probability of acquiring the skill, and the likelihood of correct or incorrect responses under the Not Learned and Learned states. The selected parameter values and their basis will be documented to support transparency and reproducibility of the learner-model configuration.

Phase 4 – BKT Implementation.

The configured parameters and knowledge-state update equations will be implemented within the SafeScape 2.0 adaptive learning mechanism. For each learner interaction, the system will use the learner's previous mastery probability and observed response to calculate the updated probability of mastery. The resulting knowledge estimate will be stored as part of the learner's knowledge vector and

subsequently made available to the Dynamic Difficulty Adjustment (DDA) component for adaptive decision-making.

Phase 5 – Algorithm Verification.

The implemented BKT procedure will undergo algorithmic verification to determine whether the system correctly implements the specified mathematical model. System-generated mastery probabilities will be compared with independently calculated expected values using predefined test cases representing correct and incorrect learner responses and different initial mastery conditions. Verification will focus on the correctness of parameter application, probability updating, boundary conditions, and consistency of computational results.

Phase 6 – Predictive Validation.

Following implementation verification, the BKT model will undergo predictive validation by examining the relationship between estimated mastery probabilities and subsequent observed learner performance. The purpose of this phase is to determine whether the knowledge estimates generated by the model provide a meaningful representation of the learner's evolving mastery of the targeted competencies.

The distinction between verification and validation is fundamental to the evaluation of the BKT component. Verification determines whether the implemented algorithm correctly conforms to its mathematical and computational specification, whereas validation examines whether the resulting learner-model estimates are sufficiently consistent with observed learner performance to support their intended use within the adaptive learning system. A computationally correct BKT implementation does not, by itself, establish that the model provides an accurate representation or useful prediction of learner mastery.

3.10.3 DDA Model Generation

The DDA model will be generated through:

1. definition of mastery thresholds;
2. mapping of mastery ranges to difficulty levels;
3. specification of scenario parameters;
4. implementation of decision rules;
5. testing using simulated mastery values;
6. logging of actual learner transitions; and
7. evaluation of adaptive effectiveness.

The resulting DDA model will use the BKT mastery estimate as its principal input.

3.10.4 LLM Evaluation Model

The LLM evaluation component will be generated through:

1. definition of emergency communication criteria;
2. creation of structured prompts;
3. definition of expected output schema;
4. implementation of API communication;
5. validation of returned JSON/structured output;
6. safety-rule validation;
7. creation of an expert-validated benchmark dataset; and
8. comparison of LLM and expert assessments.

The LLM will therefore be evaluated as an **integrated AI component**, rather than claiming that the researchers trained a new language model.

3.11 System Development Methodology

The system will be developed using an **iterative and incremental software development approach**. This approach is appropriate because SafeScape 2.0 integrates multiple interdependent components whose requirements may need refinement during implementation and testing.

The development process will consist of the following stages:

Phase 1 – Requirements Analysis

Identify:

- user requirements;
- fire-safety learning requirements;
- BKT requirements;
- DDA requirements;
- LLM/NLP requirements;
- usability requirements; and
- performance requirements.

Phase 2 – System and Data Design

Develop:

- system architecture;
- database/data structures;
- skill taxonomy;
- BKT model;
- DDA rules;
- LLM interaction workflow;
- user-interface design.

Phase 3 – Component Development

Develop independently:

- game modules;
- BKT module;
- DDA module;
- LLM/NLP module;
- scoring;
- feedback;
- persistence;
- API integration.

Phase 4 – Integration

Integrate the components into the SafeScape 2.0 application.

Phase 5 – Verification

Conduct:

- unit testing;
- integration testing;
- system testing;
- algorithm testing.

Phase 6 – Validation

Conduct:

- BKT validation;
- DDA validation;
- LLM expert validation;
- usability testing;
- educational evaluation.

Phase 7 – Refinement

Defects and usability problems identified during testing will be corrected and retested.

Phase 8 – Final Evaluation

The final version will undergo acceptance and quality evaluation before deployment.

3.11.1 Software Tools Used

The following tools will be used in the development and evaluation of SafeScape 2.0.

Tool/Technology	**Purpose**
Godot Engine	Development of the interactive game and learning environment
GDScript	Implementation of game logic, BKT, DDA, scoring, and persistence
JSON	Storage and exchange of structured learner and system data
LLM API	Natural-language processing and emergency-response evaluation
REST/API Services	Communication between the game and external AI services

Web technologies	Supporting integration and web-based deployment where applicable
Spreadsheet/Statistical Software	Data organization and statistical analysis
Version Control	Source-code version management
Browser Developer Tools	Performance and API inspection
Device Profiling Tools	FPS, memory, and performance measurement

The specific software versions and hardware specifications used during implementation and testing should be documented in the final manuscript to ensure reproducibility.

3.12 System Architecture

SafeScape 2.0 follows a modular architecture integrating the game environment, learner model, adaptive engine, conversational AI, and persistence mechanism.

The major architectural components are:

3.12.1 Presentation/Game Layer

This layer provides the learner-facing environment and includes:

- main menu;
- learner profile;
- **The Right Call**;
- **Task Master**;
- scenario interface;
- scoring;
- feedback;
- progress display; and
- adaptive scenario presentation.

3.12.2 Learner Modeling Layer

The learner modeling layer maintains the BKT knowledge vector and updates mastery estimates based on learner responses.

3.12.3 Adaptive Learning Layer

The DDA component receives the learner's current mastery state and selects an appropriate scenario difficulty.

3.12.4 NLP/LLM Layer

The LLM component processes learner-generated emergency communication and returns structured evaluation results.

3.12.5 Safety Validation Layer

LLM outputs are passed through predefined validation rules before being used for scoring or feedback. This provides a deterministic constraint layer for safety-critical criteria.

3.12.6 Persistence Layer

The persistence layer stores:

- learner progression;
- score;
- timestamp; and
- BKT knowledge vector.

3.13 Data Flow

The overall interaction can be represented as:

> **Learner → Game Interface → Scenario → Response → BKT → Mastery Estimate → DDA → Next Scenario**

For emergency communication:

> **Learner → The Right Call → Taglish Response → LLM → Structured Evaluation → Safety Validator → Score/Feedback**

For persistence:

> **Learner Interaction → Updated Knowledge State → JSON Save → Application Restart → JSON Load → Restored Knowledge State**

This architecture allows the intelligent components to operate as distinct modules while remaining integrated within the same learning environment.

Software Testing

Software testing will be conducted to determine whether SafeScape 2.0 satisfies its functional, technical, adaptive, and educational requirements. Testing will distinguish between **software verification** and **research validation**.

3.13.1 Functional Testing

Functional testing will determine whether each feature produces the expected output.

Test cases will include:

- login/profile;
- scenario selection;
- user interaction;
- scoring;
- feedback;
- timer;
- BKT update;
- DDA transition;
- LLM response processing;
- saving;
- loading; and
- progression.

The target is:

100% passing of critical functional test cases.

3.13.2 BKT Testing

BKT will be tested at two levels.

Mathematical Verification

Test cases will verify:

- correct-response update;
- incorrect-response update;
- learning transition;
- probability boundaries; and
- state persistence.

Expected values will be independently calculated and compared with system outputs.

Predictive Validation

BKT mastery estimates will be compared with independent subsequent learner-performance observations.

Depending on the form of the prediction task, metrics may include:

- accuracy;
- AUC;
- MAE;
- RMSE; and/or
- Brier score.

The selected metric will be reported together with its interpretation and evaluation procedure.

3.13.3 DDA Testing

DDA testing will verify whether each mastery state produces the correct difficulty level.

Condition	**Expected Result**
(P(L)<0.50)	Easy
(0.50\leq P(L)<0.85)	Moderate
(P(L)\geq0.85)	Hard

All predefined decision-rule test cases should produce the expected difficulty output.

DDA effectiveness will then be evaluated separately through learner-performance data.

3.13.4 LLM/NLP Testing

The LLM will be tested using a benchmark dataset containing representative emergency communication inputs, including:

- standard Taglish;
- code-switching;
- incomplete responses;
- misspellings;

- local terminology;
- ambiguous statements;
- emotionally expressed responses;
- multi-hazard situations; and
- incorrect or misleading information.

LLM outputs will be compared against expert assessments.

The following metrics may be used:

$$
Accuracy=\frac{TP+TN}{TP+TN+FP+FN}
$$

$$
Precision=\frac{TP}{TP+FP}
$$

$$
Recall=\frac{TP}{TP+FN}
$$

$$
F1=2\left(\frac{Precision\times Recall}{Precision+Recall}\right)
$$

For agreement between categorical expert and LLM classifications, **Cohen's kappa** may also be used where appropriate.

3.13.5 Learning Effectiveness Testing

The effectiveness of SafeScape 2.0 in improving fire-safety knowledge will be assessed through pre-test and post-test scores.

For each participant:

$$
d_i=y_i-x_i
$$

where:

- x_i = pre-test score;

- (y_i) = post-test score; and
- (d_i) = individual score difference.

The mean difference will be calculated as:

$$
\bar d=\frac{\sum d_i}{N}
$$

The paired-samples t-test will then be calculated as:

$$
t=\frac{\bar d}{s_d/\sqrt{N}}
$$

with:

$$
df=N-1
$$

The significance level will be set at:

$$
\alpha=0.05
$$

The null hypothesis will be rejected when the obtained (p)-value is less than 0.05. When a comparison group is used, the analysis will additionally examine whether the adaptive condition produces greater improvement than the non-adaptive condition, using an appropriate between-group analysis such as ANCOVA or an independent comparison of gain scores, depending on the final sample characteristics and assumptions.

Effect size will also be reported to determine the practical magnitude of the observed difference.

3.13.6 Usability Testing

Usability will be assessed using the **System Usability Scale (SUS)** and task-based usability testing.

Task-based testing will record:

- task completion;
- completion time;
- number of errors;
- assistance required; and
- successful completion rate.

SUS results will be reported using the standard SUS scoring procedure and interpreted using an established SUS interpretation framework.

3.13.7 User Acceptance Testing

User acceptance will be assessed using a validated five-point Likert-scale questionnaire covering:

- system acceptability;
- usability;
- content quality;
- perceived learning;
- accessibility; and
- overall usefulness.

The responses will be summarized using:

- frequency;
- percentage;
- mean; and
- standard deviation.

The previously defined target mean of **4.20 out of 5.00** may be used as the acceptance benchmark for core acceptance dimensions, provided that this criterion is specified before data analysis.

3.13.8 Performance Testing

Performance testing will evaluate:

- application loading time;

- frame rate;
- memory utilization;
- API response time;
- error rate;
- session stability; and
- recovery behavior.

For API-based operations, the study will report:

- mean latency;
- median latency;
- P95 latency; and
- failure/timeout rate.

Testing will be conducted under defined hardware, software, and network conditions.

3.13.9 Compatibility Testing

The system will be tested on the supported:

- desktop/laptop environments;
- mobile devices;
- browsers; and
- operating systems.

A compatibility matrix will document:

Device	OS	Browser	FPS	Load Time	Functionality	Result
Device 1	___	___	___	___	Pass/Fail	___
Device 2	___	___	___	___	Pass/Fail	___
Device 3	___	___	___	___	Pass/Fail	___

3.13.10 Persistence Testing

The learner-profile persistence mechanism will be tested using the following conditions:

Test ID	Condition	Expected Result
P-01	Save valid profile	Data successfully saved
P-02	Save updated BKT vector	Updated knowledge state stored

P-03	Restart application	Previous state restored
P-04	No save file	New profile initialized
P-05	Invalid JSON	Error handled safely
P-06	File-access failure	Operation fails gracefully
P-07	Multiple sessions	Latest valid state restored

The persistence mechanism will be treated as a **local data-storage mechanism**, not as a complete security or tamper-prevention mechanism.

3.13.11 Defect Classification

Testing defects will be classified according to severity.

Severity	**Description**	**Final Acceptance**
Critical	Crash, data loss, unsafe AI output, blocked progression	No unresolved critical defects
High	Major function unusable	No unresolved high defects
Medium	Function works with significant limitation	Corrected before final release where feasible
Low	Minor/cosmetic issue	May be deferred with documentation

The final system will be considered ready for deployment only after all **Critical and High** defects have been resolved and successfully retested.

3.13.14 Software Quality Evaluation

The overall technical quality of SafeScape 2.0 will be evaluated using relevant characteristics of **ISO/IEC 25010:2023**, which provides a software product quality model for specifying, measuring, and evaluating software quality. (ISO)

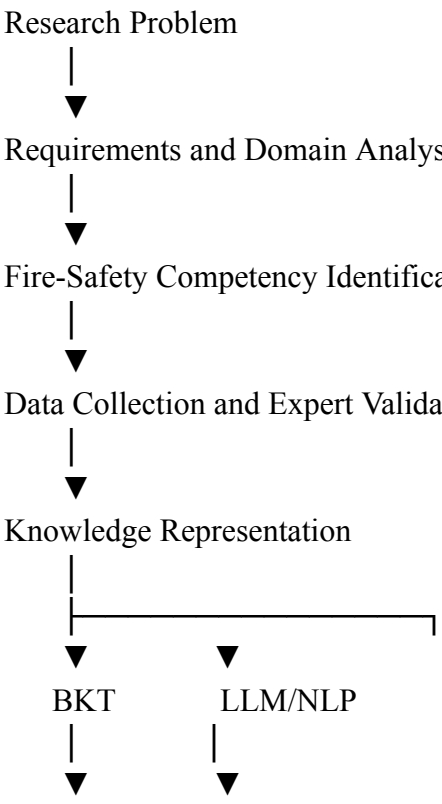
For this study, the most relevant characteristics are:

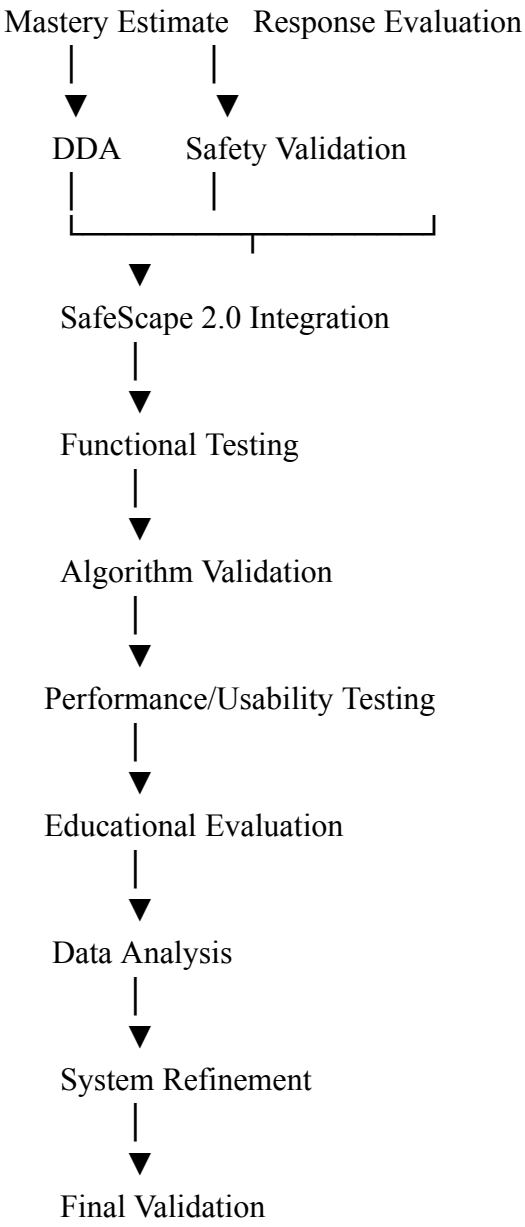
Quality Characteristic	SafeScape 2.0 Evaluation
Functional suitability	Functional test cases
Performance efficiency	FPS, latency, resource utilization
Compatibility	Device/browser testing
Usability	SUS and task-based testing
Reliability	Failure and recovery testing
Security	Access and data-handling checks
Maintainability	Modular architecture and code structure
Portability	Cross-device/browser evaluation

ISO/IEC 25010:2023 specifically provides a product-quality framework that can support requirements definition, testing objectives, quality criteria, acceptance criteria, and software evaluation. (ISO)

3.14 Overall Methodological Workflow

The complete research methodology can be summarized as follows:





The methodology combines **software development, algorithm implementation, empi**rical evaluation, and system validation to address the objectives of SafeScape 2.0. The system will be developed as an adaptive gamified intelligent tutoring environment in which BKT estimates learner mastery, DDA adjusts scenario difficulty, and an LLM/NLP component evaluates emergency communication.

The study will collect data from domain experts and target learners through expert validation, system interaction logs, knowledge assessments, LLM evaluation datasets, usability measures, and user-acceptance instruments. The computational components will be evaluated separately to distinguish implementation correctness from research validity. BKT will undergo mathematical and predictive validation, DDA will undergo rule verification and effectiveness evaluation, and the LLM will be compared with expert judgments using appropriate classification and agreement measures.

Finally, the completed system will undergo functional, integration, performance, compatibility, usability, persistence, and user-acceptance testing. Software quality evaluation will be guided by relevant characteristics of ISO/IEC 25010:2023. (ISO) This methodology is intended to provide evidence not only that SafeScape 2.0 can be successfully implemented, but also that its principal intelligent components operate correctly and contribute measurable value to the proposed fire-safety learning environment.